1. $q = n\Delta H_{vap} = \dfrac{37.6\,\text{g}}{32.042\,\text{g/mol}} \times 35.2\,\text{kJ/mol} = 41.3\,\text{kJ}$
2. Electron clouds repel strongly when they overlap ∵ at close range the filled electron clouds interpenetrate, producing a steep repulsion that strongly resists further compression.
3. It will be depressed (fall) ∵ adhesion to copper is weaker than cohesion, so the liquid is pushed down.
4. $m = \dfrac{\Delta T_b}{iK_b} = \dfrac{1.02}{1 \times 0.512} = 2.00\,\text{mol/kg}$; $n = m \times m_{\text{solvent}} = 2.00 \times 0.500\,\text{kg} = 1.00\,\text{mol}$; $m_{\text{solute}} = 1.00\,\text{mol} \times 180.16\,\text{g/mol} = 180.\,\text{g}$
5. 1 ∵ 1 F–H bond (donor) + 3 lone pairs on F (acceptor).
6. 8 ∵ center atom touches all 8 corner atoms.
7. $Xe > Kr > Ar > He$ ∵ larger molar mass → greater polarizability → stronger LDF → higher BP. Molar masses: He = 4.0 g/mol, Ar = 39.95 g/mol, Kr = 83.8 g/mol, Xe = 131.3 g/mol.
8. $\lambda = \dfrac{2d\sin\theta}{n} = \dfrac{2 \times 295 \times \sin(22.8°)}{1} = 229\,\text{pm}$. ∵ adjacent reflected waves interfere constructively only when their path difference is an integer multiple of the wavelength. The geometry gives a path difference of $2d\sin\theta$, leading to $n\lambda = 2d\sin\theta$.
9. $P_2 = P_1 \exp\left[-\dfrac{\Delta H_{vap}}{R}\left(\dfrac{1}{T_2} - \dfrac{1}{T_1}\right)\right] = 0.420\,\text{atm}$ ∵ $T_2 = 326.25\,\text{K}$, $T_1 = 353.25\,\text{K}$, $R = 8.314\,\text{J mol}^{-1}\text{K}^{-1}$.
10. $q = n\Delta H_{vap} = \dfrac{16.8\,\text{g}}{46.068\,\text{g/mol}} \times 38.6\,\text{kJ/mol} = 14.1\,\text{kJ}$
11. Subliming requires the most ∵ $\Delta H_{sub} = \Delta H_{fus} + \Delta H_{vap}$, so separating molecules straight from solid to gas exceeds either step alone.
12. $\Pi = iMRT = 1 \times 0.500 \times 0.08206 \times 298\,\text{K} = 12.2\,\text{atm}$
13. $\rho = \dfrac{Z \cdot M}{N_A \cdot a^3} = \dfrac{6 \times 47.867}{6.022 \times 10^{23} \times (295.1 \times 10^{-10})^3} = 18.6\,\text{g/mL}$
14. honey > motor oil > H_2O > CH_3COCH_3 ∵ viscosity ∝ IMF strength and molecular complexity; honey: complex mix of sugars; extensive H-bonding; motor oil: long polymer-like chains; high molecular mass; strong LDF; H_2O: H-bonding; small molecule; CH_3COCH_3: dipole-dipole; small molecule.
15. $m = \dfrac{q}{\Delta H} = \dfrac{22.7 \times 10^3\,\text{J}}{518\,\text{J/g}} = 43.8\,\text{g}$
16. Molecular solid (CO_2) ∵ molecular crystal: neutral CO_2 molecules; very low melting point due to weak LDF.
17. Warming (exothermic ΔH) favors dissolution but does not by itself prove spontaneity; both the enthalpy change and the entropy change matter. ∵ dissolving also raises entropy here, both factors are favorable and the salt dissolves spontaneously.
18. KCl: $i \times M = 2 \times 0.150\,\text{M} = 0.300\,\text{osmol/L}$ (isotonic); $C_6H_{12}O_6$: $i \times M = 1 \times 0.100\,\text{M} = 0.100\,\text{osmol/L}$ (hypotonic); Na_2SO_4: $i \times M = 3 \times 0.150\,\text{M} = 0.450\,\text{osmol/L}$ (hypertonic). ∵ tonicity depends on the total particle concentration $i \times M$, not on molarity alone.
19. CH_3OH ∴ greatest boiling point elevation ∵ largest effective molality. Na_2SO_4: $i \times m = 3 \times \dfrac{1.00}{142.04} = 0.0211\,\text{mol/kg}$; $CaCl_2$: $i \times m = 3 \times \dfrac{1.00}{111.0} = 0.0270\,\text{mol/kg}$; CH_3OH: $i \times m = 1 \times \dfrac{1.00}{32.04} = 0.0312\,\text{mol/kg}$.
20. $\rho = \dfrac{Z \cdot M}{N_A \cdot a^3} = \dfrac{2 \times 55.845}{6.022 \times 10^{23} \times (286.6 \times 10^{-10})^3} = 7.88\,\text{g/mL}$
21. 1-propanol ($CH_3CH_2CH_2OH$). ∵ Both molecules are polar, but 1-propanol can also form hydrogen bonds because it contains an O-H bond. Hydrogen bonding strengthens the attractions between molecules. Stronger intermolecular attractions make molecules harder to separate, which raises the boiling point, lowers vapour pressure, slows evaporation, and increases the energy required for vaporization.
22. $(C_2H_5)_2O$ (0.726 atm) > C_2H_5OH (0.056 atm) > H_2O (0.031 atm) ∵ stronger IMF → lower vapor pressure (H_2O: strong H-bonding; strongest IMF; lowest VP; C_2H_5OH: H-bonding; medium IMF; $(C_2H_5)_2O$: LDF only; low MM; weakest IMF).
23. $\rho = \dfrac{ZM}{N_A a^3} = \dfrac{4 \times 107.868}{6.022 \times 10^{23} \times (408.53 \times 10^{-10})^3} = 10.5\,\text{g/cm}^3$. ∵ density is determined by the mass contained in one unit cell divided by the unit-cell volume. The required information is the molar mass, the number of atoms per unit cell (obtained directly or from the crystal structure), and the unit-cell edge length; the remaining data do not affect the calculation.
24. $\Pi = iMRT = 1 \times 0.500 \times 0.08206 \times 310.\,\text{K} = 12.7\,\text{atm}$

25. Molecular solid $\because$ its composition implies a low melting point, softness, and no conduction.

26. $\Delta H_{\text{vap}} = \Delta H_{\text{sub}} - \Delta H_{\text{fus}} = 40.67 - 9.87 = 30.8\,\text{kJ/mol}$

27. $\Pi = iMRT = 2 \times 0.154 \times 0.08206 \times 298\,\text{K} = 7.54\,\text{atm}$; minimum applied pressure must exceed 7.54 atm $\because$ reverse osmosis requires $P_{\text{applied}} > \Pi$.

28. Strong electrolyte $\because$ ionic compound; dissociates completely: $Na^+ + Cl^-$.

29. $C = K_H P = 1.4 \times 10^{-3} \times 1.0\,\text{atm} = 1.4 \times 10^{-3}\,\text{mol/L}$

30. $\chi_{H_2O} = \dfrac{5.55}{5.55 + 0.899} = 0.861$; $P = 0.861 \times 149.4\,\text{mmHg} = 129\,\text{mmHg}$; $\Delta P = 20.8\,\text{mmHg}$ lowering.

31. Water flows out of the cell; the cell loses water and plasmolyzes. $\because$ the solution is hypertonic: particle concentration $= 1 \times 0.500\,\text{M} = 0.500\,\text{osmol/L} > 0.300\,\text{osmol/L}$ inside.

32. $a = \dfrac{4r}{\sqrt{3}} = \dfrac{4 \times 126}{\sqrt{3}} = 291.0\,\text{pm}$

33. Solubility increases by a factor of 5 $\because C = K_H P$; $C_1 = 6.1 \times 10^{-4}\,\text{mol/L}$, $C_2 = 3.0 \times 10^{-3}\,\text{mol/L}$; $C_2/C_1 = 5.0/1.0 = 5$.

34. Decane ($C_{10}H_{22}$). $\because$ Water forms hydrogen bonds, but decane's much larger electron cloud and greater surface area produce dispersion forces that are collectively even stronger. More energy is therefore needed to separate decane molecules. Stronger intermolecular attractions make molecules harder to separate, which raises the boiling point, lowers vapour pressure, slows evaporation, and increases the energy required for vaporization.

35. Molality (or mole fraction). $\because$ colligative formulas need a temperature-independent count of particles per solvent; molarity would drift as temperature changes the solution volume.

36. Solubility negligible change $\because$ pressure has essentially no effect on solubility of solids/liquids.

37. $P_2 = P_1 \exp\left[-\dfrac{\Delta H_{\text{vap}}}{R}\left(\dfrac{1}{T_2} - \dfrac{1}{T_1}\right)\right] = 0.533\,\text{atm} \because T_2 = 333.25\,\text{K}$, $T_1 = 353.25\,\text{K}$, $R = 8.314\,\text{J mol}^{-1}\text{K}^{-1}$.

38. Hydrogen fluoride has the higher boiling point. $\because$ its hydrogen atoms are bonded directly to fluorine, neighboring molecules can form hydrogen bonds. These intermolecular attractions are significantly stronger than the intermolecular forces in the other substance, so more energy is required to separate the molecules into the gas phase, producing the higher boiling point.

39. London dispersion and dipole–dipole $\because$ a polar molecule, but its only H is on carbon (no N/O/F–H bond).

40. $M = \dfrac{\Pi}{RT} = \dfrac{8.60 \times 10^{-3}\,\text{atm}}{0.08206 \times 298\,\text{K}} = 3.52 \times 10^{-4}\,\text{mol/L}$; $\mathscr{M} = \dfrac{m/V}{M} = \dfrac{5.00\,\text{g/L}}{3.52 \times 10^{-4}\,\text{mol/L}} \approx 1.42 \times 10^4\,\text{g/mol}$

41. $i = 2$; $m = \dfrac{0.428\,\text{mol}}{0.100\,\text{kg}} = 4.28\,\text{mol/kg}$; $\Delta T_b = i K_b m = 2 \times 0.512 \times 4.28 = 4.38\,°\text{C}$; $T_b = 100.0 + 4.38 = 104\,°\text{C}$

42. $\Pi = iMRT = 1 \times 0.0500 \times 0.08206 \times 298\,\text{K} = 1.22\,\text{atm}$; minimum applied pressure must exceed 1.22 atm $\because$ reverse osmosis requires $P_{\text{applied}} > \Pi$.

43. Both conduct electricity within the carbon sheets. $\because$ each consists of hexagonal sheets in which every carbon forms three covalent bonds, leaving electrons delocalized across the sheet that are free to move and carry charge.

44. $n_{\text{solute}} = \dfrac{25.0}{180.16} = 0.139\,\text{mol}$. $m = \dfrac{0.139\,\text{mol}}{0.500\,\text{kg}} = 0.278\,\text{mol/kg}$

45. 12 $\because$ same as FCC: 4+4+4 arrangement.

46. Ionic solid $\because$ its composition implies a high melting point, brittleness, and conduction only when molten or dissolved.

47. Sol $\because$ it is a solid dispersed in a liquid.

48. Diamond requires the much higher temperature. $\because$ diamond is a giant covalent network, so changing phase requires breaking extensive covalent bonding throughout the crystal. Solid CO_2 is a molecular solid in which only weak intermolecular attractions are overcome during the phase change.

49. $n_{eth} = \dfrac{6.40\,\text{g}}{46.07} = 0.139\,\text{mol}$; $n_{H_2O} = \dfrac{72.1\,\text{g}}{18.015} = 4.00\,\text{mol}$; $\chi_{eth} = 0.0336$; $\chi_{H_2O} = 0.966$; $\chi_{eth} + \chi_{H_2O} = 1.00$ ✓

50. $\rho = \dfrac{ZM}{N_A a^3} = \dfrac{4 \times 196.967}{6.022 \times 10^{23} \times (407.82 \times 10^{-10})^3} = 19.3\,\text{g/cm}^3$. $\because$ density is determined by the mass contained in one unit cell divided by the unit-cell volume. The required information is the molar mass, the number of atoms per unit cell (obtained directly or from the crystal structure), and the unit-cell edge length; the remaining data do not affect the calculation.

51. $n_{ace} = \dfrac{18.4\,\text{g}}{58.08} = 0.317\,\text{mol}$; $n_{H_2O} = \dfrac{100.\,\text{g}}{18.015} = 5.55\,\text{mol}$; $\chi_{ace} = 0.0540$; $\chi_{H_2O} = 0.946$; $\chi_{ace} + \chi_{H_2O} = 1.00$ ✓

52. 74.0% $\because$ 4 spheres; r = $a\sqrt{2}/4$; efficiency = $4 \times (4/3)\pi(a\sqrt{2}/4)^3/a^3 = \pi\sqrt{2}/6 \approx 74.0\%$.

53. $m = \dfrac{q}{\Delta H} = \dfrac{10.0\times 10^3\,\mathrm{J}}{571\,\mathrm{J/g}} = 17.5\,\mathrm{g}$

54. Suspension $\because$ particle size is greater than 1000 nm; large particles settle by gravity.

55. $M = \dfrac{\Pi}{RT} = \dfrac{1.90\times 10^{-3}\,\mathrm{atm}}{0.08206\times 298\,\mathrm{K}} = 7.77\times 10^{-5}\,\mathrm{mol/L}$; $\mathscr{M} = \dfrac{m/V}{M} = \dfrac{5.00\,\mathrm{g/L}}{7.77\times 10^{-5}\,\mathrm{mol/L}} \approx 6.44\times 10^4\,\mathrm{g/mol}$

56. $q = mc\Delta T = 24.2\times 2.09\times 21 = 1.06\,\mathrm{kJ}$

57. $q = n\Delta H_{\mathrm{vap}} = \dfrac{14.6\,\mathrm{g}}{32.042\,\mathrm{g/mol}}\times 35.2\,\mathrm{kJ/mol} = 16.0\,\mathrm{kJ}$

58. $CH_3NH_2 > C_2H_5NH_2 > C_3H_7NH_2 > C_6H_{13}NH_2$ $\because$ all share the same H-bonding $-NH_2$ head, so the nonpolar chain decides solubility: a longer hydrocarbon tail disrupts water's H-bond network more and lowers solubility.

59. Octahedral holes, coordination number 6 $\because$ $r_+/r_- = 72/140 = 0.51$ lies in the 0.414–0.732 range, which geometrically favors octahedral coordination.

60. 12 $\because$ each atom has 4 neighbors in its layer, 4 above, 4 below.

61. $q = n\Delta H_{\mathrm{fus}} = \dfrac{14.0\,\mathrm{g}}{78.114\,\mathrm{g/mol}}\times 9.87\,\mathrm{kJ/mol} = 1.77\,\mathrm{kJ}$

62. NaCl $\therefore$ greatest freezing point depression $\because$ largest effective molality. $C_3H_8O_3$: $i\times m = 1\times\dfrac{1.00}{92.09} = 0.0109\,\mathrm{mol/kg}$; KNO_3: $i\times m = 2\times\dfrac{1.00}{101.1} = 0.0198\,\mathrm{mol/kg}$; NaCl: $i\times m = 2\times\dfrac{1.00}{58.44} = 0.0342\,\mathrm{mol/kg}$.

63. CH_4 (-164°C) < CH_3Cl (-24°C) < CH_3SH (6°C) < CH_3OH (65°C). $\because$ Methanol forms hydrogen bonds, giving the strongest intermolecular attractions. Methanethiol and chloromethane rely primarily on dipole-dipole and London dispersion forces, while methane is nonpolar and has only weak dispersion forces.

64. $\rho = \dfrac{ZM}{N_A a^3} = \dfrac{2\times 55.845}{6.022\times 10^{23}\times(286.65\times 10^{-10})^3} = 7.87\,\mathrm{g/cm^3}$. $\because$ density is determined by the mass contained in one unit cell divided by the unit-cell volume. The required information is the molar mass, the number of atoms per unit cell (obtained directly or from the crystal structure), and the unit-cell edge length; the remaining data do not affect the calculation.

65. CH_4N_2O ($i = 1$) < $MgBr_2$ ($i = 3$) < Na_2SO_4 ($i = 3$) $\because$ all colligative properties $\propto im$; larger $i \rightarrow$ greater effect. CH_4N_2O: nonelectrolyte; 1 particle; $MgBr_2$: strong electrolyte; Mg^{2+} + $2Br^-$ $\rightarrow$ 3 particles; Na_2SO_4: strong electrolyte; $2Na^+$ + $SO_4{}^{2-}$ $\rightarrow$ 3 particles.

66. $q = n\Delta H_{\mathrm{fus}} = \dfrac{57.7\,\mathrm{g}}{55.845\,\mathrm{g/mol}}\times 13.81\,\mathrm{kJ/mol} = 14.3\,\mathrm{kJ}$

67. $n_{\mathrm{solute}} = \dfrac{25.0}{60.06} = 0.416\,\mathrm{mol}$; $n_{\mathrm{water}} = \dfrac{200.0}{18.015} = 11.1\,\mathrm{mol}$. $\chi_{\mathrm{solute}} = \dfrac{0.416}{0.416+11.1} = 0.0361$

68. $C = K_{\mathrm{H}}P = 3.4\times 10^{-2}\times 0.50\,\mathrm{atm} = 1.7\times 10^{-2}\,\mathrm{mol/L}$

69. The heat breaks intermolecular forces instead of raising kinetic energy $\because$ during a phase change energy is stored as intermolecular potential energy as solid becomes liquid, so average kinetic energy (temperature) is unchanged.

70. $\Delta H_{\mathrm{sol}} = \Delta H_{\mathrm{lattice}} + \Delta H_{\mathrm{hyd}} = +717 + (-681) = +17.2\,\mathrm{kJ/mol}$; endothermic $\because$ $\Delta H_{\mathrm{sol}} > 0$.

71. $MgCl_2$: $i\times M = 3\times 0.150\,\mathrm{M} = 0.450\,\mathrm{osmol/L}$ (hypertonic); $CaCl_2$: $i\times M = 3\times 0.100\,\mathrm{M} = 0.300\,\mathrm{osmol/L}$ (isotonic); $C_{12}H_{22}O_{11}$: $i\times M = 1\times 0.200\,\mathrm{M} = 0.200\,\mathrm{osmol/L}$ (hypotonic). $\because$ tonicity depends on the total particle concentration $i\times M$, not on molarity alone.

72. $\Delta H_{\mathrm{vap}} = 26.7\,\mathrm{kJ/mol}$ $\because$ $-R\,\dfrac{\ln(P_2/P_1)}{1/T_2 - 1/T_1}$ (Clausius–Clapeyron equation)

73. $P_2 = P_1\,\exp\left[-\dfrac{\Delta H_{\mathrm{vap}}}{R}\left(\dfrac{1}{T_2}-\dfrac{1}{T_1}\right)\right] = 0.279\,\mathrm{atm}$ $\because$ $T_2 = 320.55\,\mathrm{K}$, $T_1 = 351.55\,\mathrm{K}$, $R = 8.314\,\mathrm{J\ mol^{-1}K^{-1}}$.

74. Water has the higher boiling point. $\because$ its hydrogen atoms are bonded directly to oxygen, neighboring molecules can form hydrogen bonds. These intermolecular attractions are significantly stronger than the intermolecular forces in the other substance, so more energy is required to separate the molecules into the gas phase, producing the higher boiling point.

75. The rock salt (NaCl) structure $\because$ a face-centered cubic array of anions with cations in all of the octahedral holes is its defining arrangement.

76. No, Tyndall effect is not observed. Suspension $\because$ particles > 1000 nm; settles on standing; does not show stable Tyndall effect.

77. $m = \dfrac{q}{\Delta H} = \dfrac{14.1\times 10^3\,\mathrm{J}}{334\,\mathrm{J/g}} = 42.2\,\mathrm{g}$

78. Both have coordination number 12 and 74% packing efficiency $\because$ each atom contacts 12 neighbors; they differ only in stacking (CCP = ABCABC, HCP = ABAB).

79. A quartz crystal is crystalline (a regular, repeating lattice); rubber is amorphous (no long-range order) $\because$ crystallinity is defined by long-range periodic order.

80. It will melt $\because$ the external pressure (0.354 atm) is above the triple-point pressure (0.06 atm), so a stable liquid phase exists between solid and gas.

81. No layers form $\because$ acetic acid is polar (hydrogen-bonding) like water and hydrogen-bonds with it, so the two are miscible; density is irrelevant once the liquids mix.

82. London dispersion $\because$ nonpolar homonuclear diatomic.

83. $q = n\Delta H_{\text{fus}} = \dfrac{58.6\,\text{g}}{46.068\,\text{g/mol}} \times 4.6\,\text{kJ/mol} = 5.85\,\text{kJ}$

84. Metallic solid (Fe) $\because$ delocalized electrons (sea of electrons) allow electrical conductivity and confer malleability.

85. ABAB, repeating $\because$ in hexagonal closest packing (HCP) the third layer sits directly above the first layer, distinguishing it from the other close-packed arrangement.

86. Gas $\because$ P is below the triple-point pressure (0.117 atm), so above the triple-point temperature only gas is stable.

87. Boiling needs far more $\because$ melting only loosens the lattice (molecules stay close), whereas vaporizing must overcome nearly all intermolecular attractions to separate molecules completely.

88. $\because$ a crystal contains many families of parallel planes at different orientations, each with its own spacing d; each satisfies $n\lambda = 2d\sin\theta$ at its own angle, giving a separate peak.

89. Propan-2-ol ($(CH_3)_2CHOH$). $\because$ Both molecules have similar size, but propan-2-ol forms hydrogen bonds whereas acetone cannot hydrogen-bond to itself. The stronger attractions require more energy to overcome. Stronger intermolecular attractions make molecules harder to separate, which raises the boiling point, lowers vapour pressure, slows evaporation, and increases the energy required for vaporization.

90. It will rise $\because$ adhesion to glass exceeds cohesion, pulling the liquid up the tube.

91. $\chi_{H_2O} = \dfrac{11.1}{11.1+0.200} = 0.982$; $P = 0.982 \times 355.1\,\text{mmHg} = 349\,\text{mmHg}$; $\Delta P = 6.28\,\text{mmHg}$ lowering.

92. Positive deviation from Raoult's law $\because$ both are similar nonpolar aromatics; interactions nearly identical $\rightarrow$ ideal behavior.

93. 6 $\because$ each atom touches 4 neighbors in its plane, 1 above, 1 below.

94. N-octane $\because$ with identical formula and mass, the more elongated shape has greater surface contact area, producing stronger London dispersion forces and a higher boiling point.

95. $\Delta H_{\text{f}} = \Delta H_{\text{sub}} + IE_1 + \frac{1}{2}D + EA + U = 108 + 496 + 79 + (-328) + (-929) = -574\,\text{kJ/mol}$

96. The heat breaks intermolecular forces instead of raising kinetic energy $\because$ during a phase change energy is stored as intermolecular potential energy as solid becomes liquid, so average kinetic energy (temperature) is unchanged.

97. $\Delta H_{\text{f}} = \Delta H_{\text{sub}} + IE_1 + \frac{1}{2}D + EA + U = 108 + 496 + 79 + (-328) + (-929) = -574\,\text{kJ/mol}$

98. $a = 2\sqrt{2}r = 2\sqrt{2} \times 124 = 350.7\,\text{pm}$

99. CH_4N_2O ($i = 1$) < $MgBr_2$ ($i = 3$) < Na_2SO_4 ($i = 3$) $\because$ all colligative properties $\propto im$; larger $i \rightarrow$ greater effect. CH_4N_2O: nonelectrolyte; 1 particle; $MgBr_2$: strong electrolyte; Mg^{2+} + $2Br^-$ $\rightarrow$ 3 particles; Na_2SO_4: strong electrolyte; $2Na^+$ + $SO_4{}^{2-}$ $\rightarrow$ 3 particles.

100. $i = \dfrac{\Delta T}{K_f m} = \dfrac{0.521}{1.86 \times 0.100} = 2.80$; the solute is dissociating incompletely: the measured $i = 2.80$ falls short of the ideal 3 because ion pairing keeps some ions associated in solution.

101. $\Pi = iMRT = 1 \times 0.100 \times 0.08206 \times 298\,\text{K} = 2.45\,\text{atm}$

102. A quartz crystal is crystalline; window glass is amorphous $\because$ long-range order melts at a single temperature, while a disordered solid softens gradually as bonds of varying strength give way.

103. Vaporizing all 25.6 g needs $25.6 \times 2260 = 57.9\,\text{kJ}$, but only $31.6\,\text{kJ}$ is supplied $\therefore$ partial boil: a water–steam mixture at 100°C, with $14.0\,\text{g}$ vaporized.

104. $\Delta H_{\text{vap}} = \dfrac{-R\ln(P_2/P_1)}{1/T_2 - 1/T_1} = 26.5\,\text{kJ/mol}$

105. $P_2 = P_1\ \exp\left[-\dfrac{\Delta H_{\text{vap}}}{R}\left(\dfrac{1}{T_2} - \dfrac{1}{T_1}\right)\right] = 0.292\,\text{atm} \because T_2 = 297.15\,\text{K},\ T_1 = 329.15\,\text{K},\ R = 8.314\,\text{J mol}^{-1}\text{K}^{-1}$.

106. N-hexane $\because$ with identical formula and mass, the more elongated shape has greater surface contact area, producing stronger London dispersion forces and a higher boiling point.

107. $\because$ a crystal contains many families of parallel planes at different orientations, each with its own spacing d; each satisfies $n\lambda = 2d\sin\theta$ at its own angle, giving a separate peak.

108. Nonelectrolyte electrolyte $\because$ covalent molecular compound; does not ionize in water.

109. The substance with the deeper well $\because$ a deeper well means stronger attractions, so more thermal energy is required to disrupt its lattice and melt it.

110. The solid is denser $\because$ a positive slope means $\Delta V_{\text{fus}} < 0$, i.e. the solid is more dense than the liquid.

111. $a = \dfrac{4r}{\sqrt{3}} = \dfrac{4\times 126}{\sqrt{3}} = 291.0\,\text{pm}$

112. 4 $\because$ 8 corner atoms $\times$ 1/8 + 6 face atoms $\times$ 1/2.

113. He (760 atm) > N_2 (5.7 atm) > CO_2 (0.2 atm) > H_2O (0.031 atm) $\because$ stronger IMF $\rightarrow$ lower vapor pressure (H_2O: strong H-bonding; CO_2: LDF + weak dipole; N_2: LDF only; small; He: LDF only; tiny).

114. $a = 2\sqrt{2}r = 2\sqrt{2}\times 143 = 404.5\,\text{pm}$

115. Vaporization (liquid $\rightarrow$ gas) $\because$ the path crosses only the boiling point (-183 °C); it starts as liquid (mp -218 °C).

116. Molecular solid (CO_2) $\because$ neutral molecules held by weak IMF; low lattice energy $\rightarrow$ low melting point.

117. H_2O (0.031 atm) < CO_2 (0.2 atm) < N_2 (5.7 atm) < He (760 atm) $\because$ stronger IMF $\rightarrow$ lower vapor pressure (H_2O: strong H-bonding; CO_2: LDF + weak dipole; N_2: LDF only; small; He: LDF only; tiny).

118. Supersaturated $\because$ excess dissolved solute precipitates on the crystal nucleus.

119. Basis: 100 g solution $\rightarrow$ 25.0 g HCl, 75.0 g water; $n_{\text{solute}} = \dfrac{25.0\,\text{g}}{36.46\,\text{g/mol}} = 0.686\,\text{mol}$; $m = \dfrac{0.686\,\text{mol}}{0.0750\,\text{kg}} = 9.14\,\text{mol/kg}$

120. $\Delta H_{\text{f}} = \Delta H_{\text{sub}} + IE_1 + \frac{1}{2}D + EA + U = 108 + 496 + 79 + (-328) + (-929) = -574\,\text{kJ/mol}$

121. Molecular solid $\because$ its composition implies a low melting point, softness, and no conduction.

122. Convex (bulges up in the middle) $\because$ adhesion of mercury to glass is weaker than cohesion within the mercury, so the liquid pulls away from the walls.

123. Water flows out of the cell; the cell shrivels (crenation). $\because$ the solution is hypertonic: particle concentration $= 3\times 0.500\,\text{M} = 1.50\,\text{osmol/L} > 0.300\,\text{osmol/L}$ inside.

124. $F = 1-3+2 = 0$. $\because$ Neither temperature nor pressure can be changed independently while all three phases remain in equilibrium because the simultaneous phase equilibria completely constrain the system.

125. $\Delta H_{\text{sol}} = \Delta H_{\text{lattice}} + \Delta H_{\text{hyd}} = +853 + (-890) = -37.0\,\text{kJ/mol}$; exothermic $\because \Delta H_{\text{sol}} < 0$.

126. $m = \dfrac{0.250\,\text{mol}}{0.250\,\text{kg}} = 0.999\,\text{mol/kg}$; $\Delta T_b = iK_b m = 1\times 0.512\times 0.999 = 0.511\,°\text{C}$; $T_b = 100.0 + 0.511 = 101\,°\text{C}$

127. Boiling needs far more $\because$ melting only loosens the lattice (molecules stay close), whereas vaporizing must overcome nearly all intermolecular attractions to separate molecules completely.

128. CO_2 (-78°C) < CS_2 (46°C) < $SiCl_4$ (58°C) < CCl_4 (77°C). $\because$ All are essentially nonpolar, so London dispersion forces dominate. Larger and more polarizable electron clouds produce stronger dispersion forces and therefore higher boiling points.

129. 2 $\because$ 8 corner atoms $\times$ 1/8 + 1 body-center atom $\times$ 1.

130. NaF (-923 kJ/mol) > NaCl (-787 kJ/mol) > NaBr (-747 kJ/mol) $\because$ lattice energy becomes more negative with higher ion charges and smaller ionic radii (Coulomb's law: $U \propto \frac{q_+q_-}{r}$).

131. KF (-821 kJ/mol) < NaF (-923 kJ/mol) < LiF (-1037 kJ/mol) $\because$ lattice energy becomes more negative with higher ion charges and smaller ionic radii (Coulomb's law: $U \propto \frac{q_+q_-}{r}$).

132. The energy needed to separate the molecules completely $\because$ a deeper well means stronger intermolecular forces, so it takes more energy to pull molecules apart, giving a higher boiling point and enthalpy of vaporization.

133. $\rho = \dfrac{ZM}{N_A a^3} = \dfrac{2\times 55.845}{6.022\times 10^{23}\times(286.65\times 10^{-10})^3} = 7.87\,\text{g/cm}^3$. $\because$ density is determined by the mass contained in one unit cell divided by the unit-cell volume. The required information is the molar mass, the number of atoms per unit cell (obtained directly or from the crystal structure), and the unit-cell edge length; the remaining data do not affect the calculation.

134. $T_2 = \left(\dfrac{1}{T_1} - \dfrac{R\ln(P_2/P_1)}{\Delta H_{\text{vap}}}\right)^{-1} = 360.0\,\text{K} = 86.9°\text{C}$ $\because$ boiling occurs when the vapor pressure equals the external pressure, and a

higher external pressure raises the boiling point.

135. $\Pi = iMRT = 1 \times 0.0500 \times 0.08206 \times 298\,\text{K} = 1.22\,\text{atm}$
136. London dispersion and dipole–dipole $\because$ a polar molecule, but its only H is on carbon (no N/O/F–H bond).
137. No — it becomes a supercritical fluid with no phase boundary $\because$ above the critical temperature thermal motion prevents lasting clustering, so density rises continuously and no liquid–gas meniscus forms.
138. It is the equilibrium separation between molecules $\because$ at the minimum the net force is zero (attraction balances repulsion), and molecules in a solid vibrate about this distance.
139. Solubility increases by a factor of 4 $\because C = K_H P$; $C_1 = 6.5 \times 10^{-4}$ mol/L, $C_2 = 2.6 \times 10^{-3}$ mol/L; $C_2/C_1 = 2.0/0.5 = 4$.
140. $a = 2\sqrt{2}r = 2\sqrt{2} \times 144 = 407.3\,\text{pm}$
141. It will sublime $\because$ the external pressure (0.129 atm) is below the triple-point pressure (0.72 atm), so no liquid is stable and the solid converts directly to gas.
142. CH_4 (-164°C) < CH_3Cl (-24°C) < CH_3SH (6°C) < CH_3OH (65°C). $\because$ Methanol forms hydrogen bonds, giving the strongest intermolecular attractions. Methanethiol and chloromethane rely primarily on dipole-dipole and London dispersion forces, while methane is nonpolar and has only weak dispersion forces.
143. Strong electrolyte $\because$ ionic compound; dissociates completely: Mg^{2+} + 2 Cl^-.
144. Octahedral holes, coordination number 6 $\because r_+/r_- = 102/181 = 0.56$ lies in the 0.414–0.732 range, which geometrically favors octahedral coordination.
145. Supersaturated $\because$ excess dissolved solute precipitates on the crystal nucleus.
146. Per 1 L of solution: mass solution $= 1.02 \times 10^3$ g; mass solute $= 0.500\,\text{mol} \times 36.46\,\text{g/mol} = 18.2\,\text{g}$; mass solvent $= 1.02 \times 10^3\,\text{g} - 18.2\,\text{g} = 997\,\text{g}$; $m = \dfrac{0.500\,\text{mol}}{0.997\,\text{kg}} = 0.502\,\text{mol/kg}$
147. Exothermic ($\Delta H_{\text{sol}} = 44.5\,\text{kJ/mol}$) $\therefore$ solution temperature will increase. $\because$ solvation energy > lattice energy; system releases heat.
148. 12 $\because$ same as FCC: 4+4+4 arrangement.
149. Positive deviation from Raoult's law $\because$ hexane cannot participate in hydrogen bonding, so ethanol-hexane interactions are weaker than ethanol-ethanol interactions $\rightarrow$ higher vapor pressure.
150. Ethylene glycol $\because$ its two O–H groups plus larger size give more extensive intermolecular attraction than water; stronger intermolecular forces increase viscosity.
151. Viscosity decreases $\because$ higher temperature gives molecules more kinetic energy to overcome intermolecular attractions and slip past one another, so the liquid flows more easily.
152. $n_{ace} = \dfrac{6.40\,\text{g}}{58.08} = 0.110\,\text{mol}$; $n_{H_2O} = \dfrac{100.\,\text{g}}{18.015} = 5.55\,\text{mol}$; $\chi_{ace} = 0.0195$; $\chi_{H_2O} = 0.981$; $\chi_{ace} + \chi_{H_2O} = 1.00$ ✓
153. $P_2 = P_1 \exp\left[-\dfrac{\Delta H_{\text{vap}}}{R}\left(\dfrac{1}{T_2} - \dfrac{1}{T_1}\right)\right] = 0.418\,\text{atm}$ $\because T_2 = 315.75\,\text{K}$, $T_1 = 337.75\,\text{K}$, $R = 8.314\,\text{J mol}^{-1}\text{K}^{-1}$.
154. $\Pi = iMRT = 2 \times 0.154 \times 0.08206 \times 298\,\text{K} = 7.54\,\text{atm}$; minimum applied pressure must exceed 7.54 atm $\because$ reverse osmosis requires $P_{\text{applied}} > \Pi$.
155. $m = \dfrac{0.111\,\text{mol}}{0.100\,\text{kg}} = 1.11\,\text{mol/kg}$; $\Delta T_f = iK_f m = 1 \times 1.86 \times 1.11 = 2.06°\text{C}$; $T_f = 0.00 - 2.06 = -2.06°\text{C}$
156. Solid $\because$ at 1 atm, T is below the normal melting point (114°C).
157. H_2O (72.8 mN/m) > $C_3H_8O_3$ (63.4 mN/m) > C_6H_6 (28.9 mN/m) > C_6H_{14} (18.4 mN/m) $\because$ surface tension $\propto$ IMF strength; H_2O: strong hydrogen bonding; $C_3H_8O_3$: extensive hydrogen bonding; C_6H_6: large electron cloud; strong dispersion forces; C_6H_{14}: weaker dispersion forces.
158. $\Pi = iMRT = 2 \times 0.600 \times 0.08206 \times 298\,\text{K} = 29.4\,\text{atm}$; minimum applied pressure must exceed 29.4 atm $\because$ reverse osmosis requires $P_{\text{applied}} > \Pi$.
159. Basis: 100 g solution $\rightarrow$ 10.0 g glucose, 90.0 g water; $n_{\text{solute}} = \dfrac{10.0\,\text{g}}{180.16\,\text{g/mol}} = 0.0555\,\text{mol}$; $m = \dfrac{0.0555\,\text{mol}}{0.0900\,\text{kg}} = 0.617\,\text{mol/kg}$
160. $K_H = C/P = 6.1 \times 10^{-4}/1.0\,\text{atm} = 6.1 \times 10^{-4}\,\text{mol/(L·atm)}$; $P = \dfrac{C}{K_H} = \dfrac{1.8 \times 10^{-3}\,\text{mol/L}}{6.1 \times 10^{-4}\,\text{mol/(L·atm)}} = 3.0\,\text{atm}$
161. 52.4% $\because$ 1 sphere of radius r = a/2; sphere vol = $(4/3)\pi r^3$; cell vol = a^3; efficiency = $(4/3)\pi(a/2)^3/a^3 = \pi/6 \approx 52.4\%$.
162. Hg (485.5 mN/m) > H_2O (72.8 mN/m) > C_6H_{14} (18.4 mN/m) > CH_4 (13.0 mN/m) $\because$ surface tension $\propto$ IMF strength; Hg:

metallic bonding; extremely strong cohesion; H_2O: extensive hydrogen bonding; C_6H_{14}: London dispersion forces only; CH_4: very weak dispersion forces.

163. $K_H = C/P = 6.1\times10^{-4}/1.0\,\text{atm} = 6.1\times10^{-4}\,\text{mol/(L·atm)}$; $P = \dfrac{C}{K_H} = \dfrac{3.0\times10^{-3}\,\text{mol/L}}{6.1\times10^{-4}\,\text{mol/(L·atm)}} = 5.0\,\text{atm}$

164. $\Delta H_{vap} = 25.3\,\text{kJ/mol} \because -R\dfrac{\ln(P_2/P_1)}{1/T_2 - 1/T_1}$ (Clausius–Clapeyron equation)

165. $C = K_H P = 1.4\times10^{-3}\times0.50\,\text{atm} = 7.0\times10^{-4}\,\text{mol/L}$

166. $\Delta H_{vap} = \dfrac{38.6\,\text{kJ/mol}}{46.068\,\text{g/mol}} = 0.838\,\text{kJ/g}$

167. 68.0% $\because$ 2 spheres; r = a$\sqrt{3}$/4; efficiency = 2×(4/3)π(a$\sqrt{3}$/4)^3/a^3 = $\pi\sqrt{3}/8 \approx$ 68.0%.

168. $M = \dfrac{\Pi}{RT} = \dfrac{0.0105\,\text{atm}}{0.08206\times298\,\text{K}} = 4.29\times10^{-4}\,\text{mol/L}$; $\mathscr{M} = \dfrac{m/V}{M} = \dfrac{2.50\,\text{g/L}}{4.29\times10^{-4}\,\text{mol/L}} \approx 5.83\times10^3\,\text{g/mol}$

169. $\rho = \dfrac{Z\cdot M}{N_A\cdot a^3} = \dfrac{1\times209.0}{6.022\times10^{23}\times(335.0\times10^{-10})^3} = 9.23\,\text{g/mL}$

170. London dispersion, dipole–dipole, and hydrogen bonding $\because$ polar with N–H bonds, so all three forces act.

171. It will be depressed (fall) $\because$ adhesion to copper is weaker than cohesion, so the liquid is pushed down.

172. $\Delta H_{vap} = \dfrac{-R\ln(P_2/P_1)}{1/T_2 - 1/T_1} = 30.5\,\text{kJ/mol}$

173. Molecular solid (CO_2) $\because$ neutral molecules held by weak IMF; low lattice energy $\rightarrow$ low melting point.

174. CH_3OH $\therefore$ greatest boiling point elevation $\because$ largest effective molality. Na_2SO_4: $i\times m = 3\times\dfrac{1.00}{142.04} = 0.0211\,\text{mol/kg}$; $CaCl_2$: $i\times m = 3\times\dfrac{1.00}{111.0} = 0.0270\,\text{mol/kg}$; CH_3OH: $i\times m = 1\times\dfrac{1.00}{32.04} = 0.0312\,\text{mol/kg}$.

175. $\lambda = \dfrac{2d\sin\theta}{n} = \dfrac{2\times427\times\sin(12.1^\circ)}{1} = 179\,\text{pm}$. $\because$ adjacent reflected waves interfere constructively only when their path difference is an integer multiple of the wavelength. The geometry gives a path difference of $2d\sin\theta$, leading to $n\lambda = 2d\sin\theta$.

176. Warming the ice to 0°C uses 1.94 kJ; the remaining heat only partially melts it $\therefore$ an ice–water mixture at 0°C, with 13.7 g melted.

177. Up to 4 $\because$ 1 F–H bond (donor) + 3 lone pairs on F (acceptor).

178. $m = \dfrac{q}{\Delta H} = \dfrac{7.39\times10^3\,\text{J}}{334\,\text{J/g}} = 22.1\,\text{g}$

179. The heat breaks intermolecular forces instead of raising kinetic energy $\because$ during a phase change energy is stored as intermolecular potential energy as solid becomes liquid, so average kinetic energy (temperature) is unchanged.

180. Hydrogen bonding $\because$ water's O–H donates a hydrogen bond to acetone's carbonyl oxygen (acetone is an acceptor).

181. $m = \dfrac{\Delta T}{K_f = 1.86} = \dfrac{0.186}{1.86} = 0.100\,\text{mol/kg}$; $n = 0.100\times0.200\,\text{kg} = 0.0200\,\text{mol}$; $m_{urea} = 0.0200\,\text{mol}\times60.06\,\text{g/mol} = 1.20\,\text{g}$

182. ABAB, repeating $\because$ in hexagonal closest packing (HCP) the third layer sits directly above the first layer, distinguishing it from the other close-packed arrangement.

183. Solution A: $\Delta T = iKm = 2\times0.512\times1.00 = 1.02\,^\circ\text{C}$; Solution B: $\Delta T = 1\times0.512\times1.00 = 0.512\,^\circ\text{C}$. Solution A has the greater boiling point elevation $\because$ NaCl dissociates into 2 particles ($i = 2$) vs. glucose ($i = 1$).

184. Water flows out of the cell; the cell shrivels (crenation). $\because$ the solution is hypertonic: particle concentration $= 3\times0.150\,\text{M} = 0.450\,\text{osmol/L} > 0.300\,\text{osmol/L}$ inside.

185. Per 1 L of solution: mass solution $= 1.02\times10^3\,\text{g}$; mass solute $= 0.500\,\text{mol}\times180.16\,\text{g/mol} = 90.1\,\text{g}$; mass solvent $= 1.02\times10^3\,\text{g} - 90.1\,\text{g} = 928\,\text{g}$; $m = \dfrac{0.500\,\text{mol}}{0.928\,\text{kg}} = 0.539\,\text{mol/kg}$

186. $\Delta H_{vap} = 26.7\,\text{kJ/mol} \because -R\dfrac{\ln(P_2/P_1)}{1/T_2 - 1/T_1}$ (Clausius–Clapeyron equation)

187. Mercury $\because$ metallic bonding in mercury is far stronger than water's hydrogen bonding; stronger intermolecular forces raise surface tension.

188. Supersaturated $\because$ unstable; crystallization occurs on seeding or disturbance.

189. NaF (-923 kJ/mol) > NaCl (-787 kJ/mol) > NaBr (-747 kJ/mol) $\because$ lattice energy becomes more negative with higher ion charges and smaller ionic radii (Coulomb's law: $U \propto \dfrac{q_+q_-}{r}$).

190. $T_2 = \left(\frac{1}{T_1} - \frac{R\ln(P_2/P_1)}{\Delta H_{\text{vap}}}\right)^{-1} = 363.3\,\text{K} = 90.1°\text{C}$ ∵ boiling occurs when the vapor pressure equals the external pressure, and a lower external pressure lowers the boiling point.

191. Graphite functions better as a dry lubricant. ∵ graphite contains many stacked carbon sheets held together by weak intermolecular attractions that allow adjacent layers to slide over one another. Graphene is a single sheet and therefore lacks stacked layers that can slide.

192. Coordination $= 8 : 8$ ∵ each cation is surrounded by 8 anions and each anion by 8 cations in the cesium chloride (CsCl) structure.

193. Warming (exothermic ΔH) favors dissolution but does not by itself prove spontaneity; both the enthalpy change and the entropy change matter. ∵ dissolving also raises entropy here, both factors are favorable and the salt dissolves spontaneously.

194. He (760 atm) > N_2 (5.7 atm) > CO_2 (0.2 atm) > H_2O (0.031 atm) ∵ stronger IMF → lower vapor pressure (H_2O: strong H-bonding; CO_2: LDF + weak dipole; N_2: LDF only; small; He: LDF only; tiny).

195. $q = mc\Delta T = 11.5 \times 2.09 \times 5 = 0.120\,\text{kJ}$

196. Saturated ∵ rate of dissolution equals rate of crystallization.

197. Solution A: $\Delta T = iKm = 2 \times 1.86 \times 1.00 = 3.72\,°\text{C}$; Solution B: $\Delta T = 1 \times 1.86 \times 1.00 = 1.86\,°\text{C}$. Solution A has the greater freezing point depression ∵ NaCl dissociates into 2 particles ($i = 2$) vs. glucose ($i = 1$).

198. Hard-water Ca^{2+} and Mg^{2+} ions precipitate soap's carboxylate heads as insoluble scum, but a detergent's sulfonate/sulfate heads stay soluble with those ions, so it keeps forming micelles.

199. $C = K_H P = 1.3 \times 10^{-3} \times 2.0\,\text{atm} = 2.6 \times 10^{-3}\,\text{mol/L}$

200. $\Delta H_{\text{vap}} = \frac{40.7\,\text{kJ/mol}}{18.015\,\text{g/mol}} = 2.26\,\text{kJ/g}$

201. $Xe > Ar > Ne > He$ ∵ larger molar mass → greater polarizability → stronger LDF → higher BP. Molar masses: He = 4.0 g/mol, Ne = 20.18 g/mol, Ar = 39.95 g/mol, Xe = 131.3 g/mol.

202. He < Ne < Kr < Xe ∵ larger molar mass → greater polarizability → stronger LDF → higher BP. Molar masses: He = 4.0 g/mol, Ne = 20.18 g/mol, Kr = 83.8 g/mol, Xe = 131.3 g/mol.

203. $\Delta H_{\text{vap}} = 32.5\,\text{kJ/mol}$ ∵ $-R\frac{\ln(P_2/P_1)}{1/T_2 - 1/T_1}$ (Clausius–Clapeyron equation)

204. London dispersion and dipole–dipole ∵ the polar C–F bond gives a dipole, but the H atoms are on carbon (C–H does not hydrogen-bond).

205. $a = 2\sqrt{2}r = 2\sqrt{2} \times 124 = 350.7\,\text{pm}$

206. Solid ∵ P is below the triple-point pressure (0.06 atm) and T is below the triple-point temperature (−78°C).

207. $C_{12}H_{22}O_{11}$ ($i = 1$) < KCl ($i = 2$) < $AlCl_3$ ($i = 4$) ∵ all colligative properties $\propto im$; larger i → greater effect. $C_{12}H_{22}O_{11}$: nonelectrolyte; 1 particle; KCl: strong electrolyte; K^+ + Cl^- → 2 particles; $AlCl_3$: strong electrolyte; Al^{3+} + $3Cl^-$ → 4 particles.

208. Liquid ∵ at 1 atm, T lies between the normal melting (−210°C) and boiling (−196°C) points.

209. $Xe > Kr > Ne > He$ ∵ larger molar mass → greater polarizability → stronger LDF → higher BP. Molar masses: He = 4.0 g/mol, Ne = 20.18 g/mol, Kr = 83.8 g/mol, Xe = 131.3 g/mol.

210. $q = n\Delta H_{\text{vap}} = \frac{60.7\,\text{g}}{46.068\,\text{g/mol}} \times 38.6\,\text{kJ/mol} = 50.9\,\text{kJ}$

211. Nonelectrolyte electrolyte ∵ covalent molecular compound; does not ionize in water.

212. Concave (curves up at the walls) ∵ adhesion of kerosene to glass exceeds cohesion within the kerosene, so the liquid climbs the walls.

213. $m = \frac{0.0555\,\text{mol}}{0.100\,\text{kg}} = 0.555\,\text{mol/kg}$; $\Delta T_f = iK_f m = 1 \times 1.86 \times 0.555 = 1.03\,°\text{C}$; $T_f = 0.00 - 1.03 = -1.03\,°\text{C}$

214. $m = \frac{\Delta T_f}{K_f} = \frac{0.558}{1.86} = 0.300\,\text{mol/kg}$; $n = m \times m_{\text{solvent}} = 0.300 \times 0.200\,\text{kg} = 0.0600\,\text{mol}$; $\mathscr{M} = \frac{3.60\,\text{g}}{0.0600\,\text{mol}} = 60.1\,\text{g/mol}$

215. Negative deviation from Raoult's law ∵ extensive hydrogen bonding between methanol and water strengthens intermolecular attractions → lower vapor pressure than ideal.

216. $\Delta H_{\text{vap}} = \frac{-R\ln(P_2/P_1)}{1/T_2 - 1/T_1} = 37.7\,\text{kJ/mol}$

217. Solution A: $\Delta T = iKm = 2 \times 0.512 \times 0.100 = 0.102\,°\text{C}$; Solution B: $\Delta T = 1 \times 0.512 \times 0.100 = 0.0512\,°\text{C}$. Solution A has the greater boiling point elevation ∵ NaCl dissociates into 2 particles ($i = 2$) vs. glucose ($i = 1$).

218. $M = \frac{\rho \cdot N_A \cdot a^3}{Z} = \frac{10.51 \times 6.022 \times 10^{23} \times (408.53 \times 10^{-10})^3}{4} = 107.9\,\text{g/mol}$ ∴ the metal is silver (Ag).

219. $Xe > Kr > Ar > Ne$ ∵ larger molar mass → greater polarizability → stronger LDF → higher BP. Molar masses: Ne = 20.18 g/mol, Ar = 39.95 g/mol, Kr = 83.8 g/mol, Xe = 131.3 g/mol.

220. Solubility increases ∵ Henry's law: C = KhP; higher partial pressure of CO_2 increases solubility.

221. $M = \frac{\Pi}{RT} = \frac{0.0114\,\text{atm}}{0.08206 \times 298\,\text{K}} = 4.66 \times 10^{-4}\,\text{mol/L}$; $\mathscr{M} = \frac{m/V}{M} = \frac{30.0\,\text{g/L}}{4.66 \times 10^{-4}\,\text{mol/L}} \approx 6.44 \times 10^4\,\text{g/mol}$

222. NaF (-923 kJ/mol) > NaCl (-787 kJ/mol) > NaBr (-747 kJ/mol) ∵ lattice energy becomes more negative with higher ion charges and smaller ionic radii (Coulomb's law: $U \propto \frac{q_+ q_-}{r}$).

223. No layers form ∵ ethanol is polar (hydrogen-bonding) like water and hydrogen-bonds with it, so the two are miscible; density is irrelevant once the liquids mix.

224. Vaporization (liquid → gas) ∵ the path crosses only the boiling point (−183 °C); it starts as liquid (mp −218 °C).

225. $q = n\Delta H_{\text{vap}} = \frac{36.4\,\text{g}}{46.068\,\text{g/mol}} \times 38.6\,\text{kJ/mol} = 30.5\,\text{kJ}$

226. The liquid is denser ∵ a negative slope means $\Delta V_{\text{fus}} > 0$, i.e. the solid is less dense than the liquid.

227. 4 ∵ 8 corner atoms × 1/8 + 6 face atoms × 1/2.

228. Water (H_2O). ∵ Water molecules form an extensive hydrogen-bonding network because hydrogen is bonded to oxygen. Hydrogen sulfide cannot form comparable hydrogen bonds, so its molecules separate much more easily. Stronger intermolecular attractions make molecules harder to separate, which raises the boiling point, lowers vapour pressure, slows evaporation, and increases the energy required for vaporization.

229. The solution density. ∵ molarity is per liter of solution but molality is per kilogram of solvent; density converts the solution volume into mass so the solvent mass can be found.

230. Hydrogen bonding ∵ water's O–H donates a hydrogen bond to acetone's carbonyl oxygen (acetone is an acceptor).

231. N-pentane ∵ with identical formula and mass, the more elongated shape has greater surface contact area, producing stronger London dispersion forces and a higher boiling point.

232. Hg (485.5 mN/m) > H_2O (72.8 mN/m) > C_6H_6 (28.9 mN/m) > $(C_2H_5)_2O$ (17.0 mN/m) ∵ surface tension ∝ IMF strength; Hg: metallic bonding; H_2O: strong hydrogen bonding; C_6H_6: strong dispersion forces; $(C_2H_5)_2O$: weak intermolecular attractions.

233. CH_4N_2O ($i = 1$) < $MgBr_2$ ($i = 3$) < Na_2SO_4 ($i = 3$) ∵ all colligative properties ∝ im; larger i → greater effect. CH_4N_2O: nonelectrolyte; 1 particle; $MgBr_2$: strong electrolyte; Mg^{2+} + $2Br^-$ → 3 particles; Na_2SO_4: strong electrolyte; $2Na^+$ + $SO_4{}^{2-}$ → 3 particles.

234. $\Delta H_{\text{sub}} = \Delta H_{\text{fus}} + \Delta H_{\text{vap}} = 6.01 + 40.7 = 46.71\,\text{kJ/mol}$

235. $\Delta H_{\text{vap}} = 1099\,\text{J/g} \times 32.042\,\text{g/mol} \times \frac{1\,\text{kJ}}{1000\,\text{J}} = 35.2\,\text{kJ/mol}$

236. LiBr ∴ greatest freezing point depression ∵ largest effective molality. $MgSO_4$: $i \times m = 2 \times \frac{1.00}{120.37} = 0.0166\,\text{mol/kg}$; C_2H_5OH: $i \times m = 1 \times \frac{1.00}{46.07} = 0.0217\,\text{mol/kg}$; LiBr: $i \times m = 2 \times \frac{1.00}{86.85} = 0.0230\,\text{mol/kg}$.

237. Octane (C_8H_{18}). ∵ Although ethanol can hydrogen-bond, octane has a much larger and more polarizable electron cloud. The stronger cumulative London dispersion forces require more energy to separate its molecules. Stronger intermolecular attractions make molecules harder to separate, which raises the boiling point, lowers vapour pressure, slows evaporation, and increases the energy required for vaporization.

238. $\Delta H_{\text{vap}} = \frac{30.8\,\text{kJ/mol}}{78.114\,\text{g/mol}} = 0.394\,\text{kJ/g}$

239. hexagonal closest packing (HCP), coordination number 12 ∵ the ABAB repeat is its signature; each atom touches 6 in its layer plus 3 above and 3 below.

240. $m = \frac{q}{\Delta H} = \frac{36.6 \times 10^3\,\text{J}}{247\,\text{J/g}} = 148\,\text{g}$

241. London dispersion, dipole–dipole, and hydrogen bonding ∵ polar with N–H bonds, so all three forces act.

242. $\Delta H_{\text{vap}} = 83.5\,\text{kJ/mol}$ ∵ $-R\frac{\ln(P_2/P_1)}{1/T_2 - 1/T_1}$ (Clausius–Clapeyron equation)

243. $m = \frac{\Delta T_f}{K_f} = \frac{1.86}{37.7} = 0.0493\,\text{mol/kg}$; $n = m \times m_{\text{solvent}} = 0.0493 \times 0.250\,\text{kg} = 0.0123\,\text{mol}$; $\mathscr{M} = \frac{3.03\,\text{g}}{0.0123\,\text{mol}} = 246\,\text{g/mol}$

244. $F = 1 - 2 + 2 = 1$. $\because$ Only one intensive variable may be chosen independently because maintaining equilibrium between two phases imposes one independent constraint relating temperature and pressure.

245. Increasing pressure raises the melting point $\because$ pressure favors the denser solid (smaller molar volume), so the solid–liquid boundary has a positive slope.

246. Warming (exothermic ΔH) favors dissolution but does not by itself prove spontaneity; both the enthalpy change and the entropy change matter. $\because$ dissolving also raises entropy here, both factors are favorable and the salt dissolves spontaneously.

247. $m = \dfrac{\Delta T}{K_f = 1.86} = \dfrac{1.02}{1.86} = 0.551\,\text{mol/kg}$; $n = 0.551 \times 0.200\,\text{kg} = 0.110\,\text{mol}$; $m_{anth} = 0.110\,\text{mol} \times 178.23\,\text{g/mol} = 19.6\,\text{g}$

248. $\Delta H_{\text{vap}} = \Delta H_{\text{sub}} - \Delta H_{\text{fus}} = 57.09 - 15.52 = 41.57\,\text{kJ/mol}$

249. $\Delta H_{\text{sol}} = \Delta H_{\text{lattice}} + \Delta H_{\text{hyd}} = +929 + (-904) = +1.9\,\text{kJ/mol}$; endothermic $\because \Delta H_{\text{sol}} > 0$.

250. $m = \dfrac{\Delta T_f}{K_f} = \dfrac{0.512}{1.86} = 0.275\,\text{mol/kg}$; $n = m \times m_{\text{solvent}} = 0.275 \times 0.250\,\text{kg} = 0.0688\,\text{mol}$; $\mathscr{M} = \dfrac{23.6\,\text{g}}{0.0688\,\text{mol}} = 342\,\text{g/mol}$

251. Sodium chloride is crystalline; window glass is amorphous $\because$ long-range order melts at a single temperature, while a disordered solid softens gradually as bonds of varying strength give way.

252. True solution $\because$ particle size is less than 1 nm; dissolved species are molecular or ionic in size.

253. Solubility increases by a factor of 5 $\because C = K_{\text{H}}P$; $C_1 = 7.8 \times 10^{-4}\,\text{mol/L}$, $C_2 = 3.9 \times 10^{-3}\,\text{mol/L}$; $C_2/C_1 = 5.0/1.0 = 5$.

254. CH_4N_2O: $i \times M = 1 \times 0.150\,\text{M} = 0.150\,\text{osmol/L}$ (hypotonic); Na_2SO_4: $i \times M = 3 \times 0.150\,\text{M} = 0.450\,\text{osmol/L}$ (hypertonic); NaCl: $i \times M = 2 \times 0.150\,\text{M} = 0.300\,\text{osmol/L}$ (isotonic). $\because$ tonicity depends on the total particle concentration $i \times M$, not on molarity alone.

255. Water flows into the cell; the cell takes up water and becomes turgid. $\because$ the solution is hypotonic: particle concentration $= 3 \times 0.0500\,\text{M} = 0.150\,\text{osmol/L} < 0.300\,\text{osmol/L}$ inside.

256. Kno_3 crystallizes from solution on cooling. $\because$ KNO_3 solubility increases with T; cooling creates supersaturation $\rightarrow$ crystallization.

257. $q = m\Delta H_{\text{vap}} = 30.7 \times 2260 = 69.4\,\text{kJ}$

258. $m = \dfrac{\Delta T}{K_b = 0.512} = \dfrac{0.512}{0.512} = 1.00\,\text{mol/kg}$; $n = 1.00 \times 0.200\,\text{kg} = 0.200\,\text{mol}$; $m_{naph} = 0.200\,\text{mol} \times 128.17\,\text{g/mol} = 25.6\,\text{g}$

259. Vaporization (liquid $\rightarrow$ gas) $\because$ the path crosses only the boiling point (-183 °C); it starts as liquid (mp -218 °C).

260. $\lambda = \dfrac{2d\sin\theta}{n} = \dfrac{2 \times 196 \times \sin(23.1°)}{1} = 154\,\text{pm}$. $\because$ adjacent reflected waves interfere constructively only when their path difference is an integer multiple of the wavelength. The geometry gives a path difference of $2d\sin\theta$, leading to $n\lambda = 2d\sin\theta$.

261. HCOOH > CH_3COOH > C_2H_5COOH > C_3H_7COOH $\because$ all share the same H-bonding $-$COOH head, so the nonpolar chain decides solubility: a longer hydrocarbon tail disrupts water's H-bond network more and lowers solubility.

262. $\Delta H_{\text{vap}} = \Delta H_{\text{sub}} - \Delta H_{\text{fus}} = 46.71 - 6.01 = 40.7\,\text{kJ/mol}$

263. Suspension $\because$ particle size is greater than 1000 nm; large particles settle by gravity.

264. $M = \dfrac{\Pi}{RT} = \dfrac{0.0171\,\text{atm}}{0.08206 \times 298\,\text{K}} = 6.99 \times 10^{-4}\,\text{mol/L}$; $\mathscr{M} = \dfrac{m/V}{M} = \dfrac{10.0\,\text{g/L}}{6.99 \times 10^{-4}\,\text{mol/L}} \approx 1.43 \times 10^4\,\text{g/mol}$

265. $m = \dfrac{\Delta T}{K_f = 1.86} = \dfrac{0.186}{1.86} = 0.100\,\text{mol/kg}$; $n = 0.100 \times 0.100\,\text{kg} = 0.0100\,\text{mol}$; $m_{urea} = 0.0100\,\text{mol} \times 60.06\,\text{g/mol} = 0.601\,\text{g}$

266. $F = 1 - 2 + 2 = 1$. $\because$ Only one intensive variable may be chosen independently because maintaining equilibrium between two phases imposes one independent constraint relating temperature and pressure.

267. No layers form $\because$ acetic acid is polar (hydrogen-bonding) like water and hydrogen-bonds with it, so the two are miscible; density is irrelevant once the liquids mix.

268. Graphite conducts more effectively. $\because$ each carbon atom in graphite forms three covalent bonds, leaving electrons delocalized across each sheet that can move and carry charge. In diamond every carbon forms four localized covalent bonds in a three-dimensional network, leaving no mobile electrons.

269. Yes $\because$ ion-dipole forces between Na^+/Cl^- and H_2O overcome the lattice energy.

270. Solubility decreases $\because$ gas solubility always decreases with increasing temperature.

271. Hydrogen bonding $\because$ O–H group allows H-bonding.

272. Negative slope $\because$ the solid is less dense than the liquid ($\Delta V_{\text{fus}} > 0$), so dP/dT is negative.

273. He (760 atm) > N_2 (5.7 atm) > CO_2 (0.2 atm) > H_2O (0.031 atm) $\because$ stronger IMF $\rightarrow$ lower vapor pressure (H_2O: strong

H-bonding; CO_2: LDF + weak dipole; N_2: LDF only; small; He: LDF only; tiny).

274. No — spontaneity is not the same as exothermicity. $\because$ dissolving scatters the ordered crystal lattice into freely moving hydrated ions, the large entropy increase outweighs the endothermic ΔH, so the process proceeds even while it cools the surroundings.

275. Per 1 L of solution: mass solution $= 1.02 \times 10^3$ g; mass solute $= 0.500\,\text{mol} \times 36.46\,\text{g/mol} = 18.2\,\text{g}$; mass solvent $= 1.02 \times 10^3\,\text{g} - 18.2\,\text{g} = 997\,\text{g}$; $m = \dfrac{0.500\,\text{mol}}{0.997\,\text{kg}} = 0.502\,\text{mol/kg}$

276. Gas $\because$ P is below the triple-point pressure (0.124 atm), so above the triple-point temperature only gas is stable.

277. ABCABC, repeating $\because$ in cubic closest packing (CCP) the third layer occupies new positions before the pattern repeats, distinguishing it from the other close-packed arrangement.

278. $\because$ a crystal contains many families of parallel planes at different orientations, each with its own spacing d; each satisfies $n\lambda = 2d\sin\theta$ at its own angle, giving a separate peak.

279. Solubility increases by a factor of 2 $\because C = K_H P$; $C_1 = 1.6 \times 10^{-3}$ mol/L, $C_2 = 3.1 \times 10^{-3}$ mol/L; $C_2/C_1 = 4.0/2.0 = 2$.

280. Below it the molecules crowd the surface and stay dispersed; once the water is saturated with single molecules, adding more drives them to cluster tails-inward as micelles.

281. $C_2H_5NH_2 > C_3H_7NH_2 > C_6H_{13}NH_2 > C_8H_{17}NH_2$ $\because$ all share the same H-bonding $-NH_2$ head, so the nonpolar chain decides solubility: a longer hydrocarbon tail disrupts water's H-bond network more and lowers solubility.

282. $Xe > Kr > Ne > He$ $\because$ larger molar mass $\rightarrow$ greater polarizability $\rightarrow$ stronger LDF $\rightarrow$ higher BP. Molar masses: He = 4.0 g/mol, Ne = 20.18 g/mol, Kr = 83.8 g/mol, Xe = 131.3 g/mol.

283. It dissolves caffeine yet penetrates and evaporates cleanly $\because$ liquid-like density dissolves caffeine while gas-like low viscosity penetrates the beans; releasing pressure evaporates it, leaving no residue.

284. $m = \dfrac{\Delta T_f}{K_f} = \dfrac{1.86}{5.12} = 0.363\,\text{mol/kg}$; $n = m \times m_{\text{solvent}} = 0.363 \times 0.250\,\text{kg} = 0.0908\,\text{mol}$; $\mathscr{M} = \dfrac{11.1\,\text{g}}{0.0908\,\text{mol}} = 122\,\text{g/mol}$

285. $\Delta H_{\text{vap}} = 25.3\,\text{kJ/mol} \because -R\,\dfrac{\ln(P_2/P_1)}{1/T_2 - 1/T_1}$ (Clausius–Clapeyron equation)

286. Suspension $\because$ particle size is greater than 1000 nm; suspensions are heterogeneous mixtures.

287. Co_2 solubility decreases; gas escapes. $\because$ gas solubility decreases with temperature; higher T $\rightarrow$ lower CO_2 dissolved $\rightarrow$ flat soda.

288. $q = n\Delta H_{\text{fus}} = \dfrac{49.3\,\text{g}}{46.068\,\text{g/mol}} \times 4.6\,\text{kJ/mol} = 4.92\,\text{kJ}$

289. Both conduct electricity within the carbon sheets. $\because$ each consists of hexagonal sheets in which every carbon forms three covalent bonds, leaving electrons delocalized across the sheet that are free to move and carry charge.

290. $m = \dfrac{\Delta T_f}{K_f} = \dfrac{1.86}{5.12} = 0.363\,\text{mol/kg}$; $n = m \times m_{\text{solvent}} = 0.363 \times 0.250\,\text{kg} = 0.0908\,\text{mol}$; $\mathscr{M} = \dfrac{7.09\,\text{g}}{0.0908\,\text{mol}} = 78.1\,\text{g/mol}$

291. $F = 1 - 2 + 2 = 1$. $\because$ Only one intensive variable may be chosen independently because maintaining equilibrium between two phases imposes one independent constraint relating temperature and pressure.

292. No, Tyndall effect is not observed. True solution $\because$ Na^+ and Cl^- ions are < 1 nm; too small to scatter light; no Tyndall effect.

293. The solution density. $\because$ molarity is per liter of solution but molality is per kilogram of solvent; density converts the solution volume into mass so the solvent mass can be found.

294. $F = 1 - 2 + 2 = 1$. $\because$ Only one intensive variable may be chosen independently because maintaining equilibrium between two phases imposes one independent constraint relating temperature and pressure.

295. $q = q_1 + q_2 + q_3 + q_4 + q_5 = m(c_{\text{ice}} \cdot 15 + \Delta H_{\text{fus}} + c_{\text{water}} \cdot 100 + \Delta H_{\text{vap}} + c_{\text{steam}} \cdot 26) = 78.6\,\text{kJ}$

296. Subliming requires the most $\because$ $\Delta H_{\text{sub}} = \Delta H_{\text{fus}} + \Delta H_{\text{vap}}$, so separating molecules straight from solid to gas exceeds either step alone.

297. Yes $\because$ ion-dipole forces between Na^+/Cl^- and H_2O overcome the lattice energy.

298. The solution density. $\because$ molarity is per liter of solution but molality is per kilogram of solvent; density converts the solution volume into mass so the solvent mass can be found.

299. Two layers form and water is on top $\because$ chloroform is nonpolar and cannot hydrogen-bond with water, so they are immiscible; the less dense liquid floats, and density fixes only the stacking order, not whether they mix.

300. Solubility increases slightly $\because$ dissolution of NaCl is slightly endothermic; higher T slightly favors dissolution.

301. Weak electrolyte $\because$ weak acid; only partially ionizes (Ka = 1.8×10^{-5}); mostly undissociated in solution.

302. $m = \dfrac{\Delta T}{K_b = 0.512} = \dfrac{1.02}{0.512} = 2.00\,\text{mol/kg}$; $n = 2.00 \times 0.500\,\text{kg} = 1.00\,\text{mol}$; $m_{anth} = 1.00\,\text{mol} \times 178.23\,\text{g/mol} = 178\,\text{g}$

303. $n_{\text{solute}} = \dfrac{20.0}{62.07} = 0.322\,\text{mol}$; $n_{\text{water}} = \dfrac{100.0}{18.015} = 5.55\,\text{mol}$. $\chi_{\text{solute}} = \dfrac{0.322}{0.322+5.55} = 0.0549$

304. It dissolves caffeine yet penetrates and evaporates cleanly ∵ liquid-like density dissolves caffeine while gas-like low viscosity penetrates the beans; releasing pressure evaporates it, leaving no residue.

305. No ∵ the lattice energy of AgCl is too large for water to effectively separate the ions.

306. Basis: 100 g solution → $5.00\,\text{g}$ KCl, $95.0\,\text{g}$ water; $n_{\text{solute}} = \dfrac{5.00\,\text{g}}{74.55\,\text{g/mol}} = 0.0671\,\text{mol}$; $m = \dfrac{0.0671\,\text{mol}}{0.0950\,\text{kg}} = 0.706\,\text{mol/kg}$

307. True solution ∵ particle size is less than 1 nm; dissolved species are molecular or ionic in size.

308. Viscosity decreases ∵ higher temperature gives molecules more kinetic energy to overcome intermolecular attractions and slip past one another, so the liquid flows more easily.

309. $\lambda = \dfrac{2d\sin\theta}{n} = \dfrac{2 \times 488 \times \sin(28.0°)}{2} = 229\,\text{pm}$. ∵ adjacent reflected waves interfere constructively only when their path difference is an integer multiple of the wavelength. The geometry gives a path difference of $2d\sin\theta$, leading to $n\lambda = 2d\sin\theta$.

310. $CaCl_2$: $i \times M = 3 \times 0.0500\,\text{M} = 0.150\,\text{osmol/L}$ (hypotonic); $MgCl_2$: $i \times M = 3 \times 0.100\,\text{M} = 0.300\,\text{osmol/L}$ (isotonic); CH_4N_2O: $i \times M = 1 \times 0.250\,\text{M} = 0.250\,\text{osmol/L}$ (hypotonic). ∵ tonicity depends on the total particle concentration $i \times M$, not on molarity alone.

311. H_2O (72.8 mN/m) > $HOCH_2CH_2OH$ (47.7 mN/m) > CH_3COCH_3 (23.7 mN/m) > C_6H_{14} (18.4 mN/m) ∵ surface tension ∝ IMF strength; H_2O: strong hydrogen bonding; $HOCH_2CH_2OH$: multiple hydrogen-bonding sites; CH_3COCH_3: dipole-dipole interactions; C_6H_{14}: London dispersion forces only.

312. Water flows into the cell; the cell swells and may lyse. ∵ the solution is hypotonic: particle concentration $= 3 \times 0.0500\,\text{M} = 0.150\,\text{osmol/L} < 0.300\,\text{osmol/L}$ inside.

313. N-pentane ∵ with identical formula and mass, the more elongated shape has greater surface contact area, producing stronger London dispersion forces and a higher boiling point.

314. corn syrup > olive oil > H_2O > $C_2H_5OC_2H_5$ ∵ viscosity ∝ IMF strength and molecular complexity; corn syrup: high sugar concentration with extensive H-bonding; olive oil: large hydrocarbon chains produce stronger LDF; H_2O: small but hydrogen-bonded; $C_2H_5OC_2H_5$: weak dipole forces and limited H-bonding.

315. Diethyl ether ($C_2H_5OC_2H_5$). ∵ Diethyl ether is polar and experiences dipole-dipole attractions in addition to London dispersion forces. Pentane relies only on dispersion forces despite having similar size. Stronger intermolecular attractions make molecules harder to separate, which raises the boiling point, lowers vapour pressure, slows evaporation, and increases the energy required for vaporization.

316. Positive deviation from Raoult's law ∵ hexane cannot participate in hydrogen bonding, so ethanol-hexane interactions are weaker than ethanol-ethanol interactions → higher vapor pressure.

317. $m = \dfrac{0.333\,\text{mol}}{0.100\,\text{kg}} = 3.33\,\text{mol/kg}$; $\Delta T_b = iK_b m = 1 \times 0.512 \times 3.33 = 1.70\,°\text{C}$; $T_b = 100.0 + 1.70 = 102\,°\text{C}$

318. H_2O (72.8 mN/m) > $HOCH_2CH_2OH$ (47.7 mN/m) > CH_3COCH_3 (23.7 mN/m) > C_6H_{14} (18.4 mN/m) ∵ surface tension ∝ IMF strength; H_2O: strong hydrogen bonding; $HOCH_2CH_2OH$: multiple hydrogen-bonding sites; CH_3COCH_3: dipole-dipole interactions; C_6H_{14}: London dispersion forces only.

319. N-octane ∵ with identical formula and mass, the more elongated shape has greater surface contact area, producing stronger London dispersion forces and a higher boiling point.

320. The three phases have different specific heat capacities ∵ slope is inversely related to specific heat ($q = mc\Delta T$), so a phase with larger c warms more slowly and has a gentler slope.

321. Melting all 52.8 g needs $52.8 \times 334 = 17.6\,\text{kJ}$, but only $13.9\,\text{kJ}$ is supplied ∴ partial melt: an ice–water mixture at 0°C, with $41.6\,\text{g}$ melted.

322. H_2O (0.031 atm) < CO_2 (0.2 atm) < N_2 (5.7 atm) < He (760 atm) ∵ stronger IMF → lower vapor pressure (H_2O: strong H-bonding; CO_2: LDF + weak dipole; N_2: LDF only; small; He: LDF only; tiny).

323. Ammonia has the higher boiling point. ∵ its hydrogen atoms are bonded directly to nitrogen, neighboring molecules can form hydrogen bonds. These intermolecular attractions are significantly stronger than the intermolecular forces in the other substance, so more energy is required to separate the molecules into the gas phase, producing the higher boiling point.

324. CO_2 (-78°C) < CS_2 (46°C) < $SiCl_4$ (58°C) < CCl_4 (77°C). ∵ All are essentially nonpolar, so London dispersion forces dominate. Larger and more polarizable electron clouds produce stronger dispersion forces and therefore higher boiling points.

325. No — spontaneity is not the same as exothermicity. $\because$ dissolving scatters the ordered crystal lattice into freely moving hydrated ions, the large entropy increase outweighs the endothermic ΔH, so the process proceeds even while it cools the surroundings.

326. $P_2 = P_1 \exp\left[-\frac{\Delta H_{\text{vap}}}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)\right] = 0.555\,\text{atm} \because T_2 = 336.55\,\text{K}, T_1 = 351.55\,\text{K}, R = 8.314\,\text{J mol}^{-1}\text{K}^{-1}$.

327. $\Pi = iMRT = 1 \times 0.100 \times 0.08206 \times 298\,\text{K} = 2.45\,\text{atm}$

328. X-rays are required $\because$ diffraction needs a wavelength comparable to the plane spacing ($\sim$100 pm); X-ray wavelengths match atomic spacings, while visible light ($\sim$500 nm) is far too long to resolve them.

329. Per 1 L of solution: mass solution $= 1.05 \times 10^3$ g; mass solute $= 1.00\,\text{mol} \times 58.44\,\text{g/mol} = 58.4\,\text{g}$; mass solvent $= 1.05 \times 10^3\,\text{g} - 58.4\,\text{g} = 992\,\text{g}$; $m = \frac{1.00\,\text{mol}}{0.992\,\text{kg}} = 1.01\,\text{mol/kg}$

330. Sodium chloride is crystalline (a regular, repeating lattice); window glass is amorphous (no long-range order) $\because$ crystallinity is defined by long-range periodic order.

331. castor oil > $HOCH_2CH_2OH$ > H_2O > CH_3COCH_3 $\because$ viscosity $\propto$ IMF strength and molecular complexity; castor oil: large triglycerides with strong dispersion forces; $HOCH_2CH_2OH$: strong H-bonding from two alcohol groups; H_2O: hydrogen bonding but low molar mass; CH_3COCH_3: small polar molecule with weaker IMF.

332. $\Delta H_{\text{vap}} = \frac{31.3\,\text{kJ/mol}}{58.08\,\text{g/mol}} = 0.539\,\text{kJ/g}$

333. Hard-water Ca^{2+} and Mg^{2+} ions precipitate soap's carboxylate heads as insoluble scum, but a detergent's sulfonate/sulfate heads stay soluble with those ions, so it keeps forming micelles.

334. $m = \frac{\Delta T_b}{iK_b} = \frac{0.256}{1 \times 0.512} = 0.500\,\text{mol/kg}$; $n = m \times m_{\text{solvent}} = 0.500 \times 0.200\,\text{kg} = 0.100\,\text{mol}$; $m_{\text{solute}} = 0.100\,\text{mol} \times 180.16\,\text{g/mol} = 18.0\,\text{g}$

335. Metallic solid (Cu) $\because$ metallic crystal: delocalized electrons carry charge at all temperatures.

336. London dispersion and dipole–dipole $\because$ the polar C–F bond gives a dipole, but the H atoms are on carbon (C–H does not hydrogen-bond).

337. The amphiphilic molecules bury their nonpolar tails in the grease while their ionic heads face the water, forming micelles that suspend the grease as a colloidal dispersion that rinses away.

338. Covalent-network solid (diamond (C)) $\because$ atoms bonded in extended covalent network; no free charges.

339. $m = \frac{\Delta T_b}{iK_b} = \frac{0.256}{1 \times 0.512} = 0.500\,\text{mol/kg}$; $n = m \times m_{\text{solvent}} = 0.500 \times 0.200\,\text{kg} = 0.100\,\text{mol}$; $m_{\text{solute}} = 0.100\,\text{mol} \times 60.06\,\text{g/mol} = 6.01\,\text{g}$

340. $\rho = \frac{Z \cdot M}{N_A \cdot a^3} = \frac{4 \times 97.46}{6.022 \times 10^{23} \times (541.0 \times 10^{-10})^3} = 4.09\,\text{g/mL}$

341. It will sublime $\because$ the external pressure (0.3612 atm) is below the triple-point pressure (0.72 atm), so no liquid is stable and the solid converts directly to gas.

342. $\chi_{H_2O} = \frac{9.99}{9.99 + 0.300} = 0.971$; $P = 0.971 \times 23.8\,\text{mmHg} = 23.1\,\text{mmHg}$; $\Delta P = 0.693\,\text{mmHg}$ lowering.

343. London dispersion forces $\because$ both are nonpolar, so only induced-dipole (dispersion) forces act between them.

344. Dissolved o_2 decreases. $\because$ gas solubility decreases with increasing temperature; fish kills can result.

345. Metallic solid (Cu) $\because$ metallic crystal: delocalized electrons carry charge at all temperatures.

346. $n_{\text{solute}} = \frac{25.0}{342.3} = 0.0730\,\text{mol}$. $m = \frac{0.0730\,\text{mol}}{0.250\,\text{kg}} = 0.292\,\text{mol/kg}$

347. The fluorite (CaF2) structure $\because$ a face-centered cubic array of cations with anions in all of the tetrahedral holes is its defining arrangement.

348. $m = \frac{\Delta T}{K_f = 1.86} = \frac{0.186}{1.86} = 0.100\,\text{mol/kg}$; $n = 0.100 \times 0.200\,\text{kg} = 0.0200\,\text{mol}$; $m_{naph} = 0.0200\,\text{mol} \times 128.17\,\text{g/mol} = 2.56\,\text{g}$

349. 2 $\because$ 1 O–H bond (donor) + 2 lone pairs on carbonyl/hydroxyl O (acceptor).

350. No $\because$ weak LDF of oil cannot compete with strong H-bonding network of water.

351. Yes $\because$ a reflection needs $\sin\theta = \lambda/2d \leq 1$, i.e. $\lambda \leq 2d = 784$ pm; here $\lambda = 194$ pm satisfies this.

352. $i = \frac{\Delta T}{K_f m} = \frac{0.335}{1.86 \times 0.100} = 1.80$; the solute is dissociating incompletely: the measured $i = 1.80$ falls short of the ideal 2 because ion pairing keeps some ions associated in solution.

353. Negative slope $\because$ the solid is less dense than the liquid ($\Delta V_{\text{fus}} > 0$), so dP/dT is negative.

354. $\rho = \dfrac{ZM}{N_A a^3} = \dfrac{4 \times 26.982}{6.022 \times 10^{23} \times (404.95 \times 10^{-10})^3} = 2.70\,\text{g/cm}^3$. $\because$ density is determined by the mass contained in one unit cell divided by the unit-cell volume. The required information is the molar mass, the number of atoms per unit cell (obtained directly or from the crystal structure), and the unit-cell edge length; the remaining data do not affect the calculation.

355. $\Delta H_{\text{vap}} = \dfrac{-R\ln(P_2/P_1)}{1/T_2 - 1/T_1} = 26.5\,\text{kJ/mol}$

356. $m = \dfrac{\Delta T_f}{K_f} = \dfrac{0.372}{5.12} = 0.0727\,\text{mol/kg}$; $n = m \times m_{\text{solvent}} = 0.0727 \times 0.200\,\text{kg} = 0.0145\,\text{mol}$; $\mathscr{M} = \dfrac{1.77\,\text{g}}{0.0145\,\text{mol}} = 122\,\text{g/mol}$

357. Solution A: $\Delta T = iKm = 2 \times 0.512 \times 1.00 = 1.02\,°\text{C}$; Solution B: $\Delta T = 1 \times 0.512 \times 1.00 = 0.512\,°\text{C}$. Solution A has the greater boiling point elevation $\because$ NaCl dissociates into 2 particles ($i = 2$) vs. glucose ($i = 1$).

358. $K_{\text{H}} = C/P = 6.1 \times 10^{-4}/1.0\,\text{atm} = 6.1 \times 10^{-4}\,\text{mol/(L·atm)}$; $P = \dfrac{C}{K_{\text{H}}} = \dfrac{1.2 \times 10^{-3}\,\text{mol/L}}{6.1 \times 10^{-4}\,\text{mol/(L·atm)}} = 2.0\,\text{atm}$

359. Boiling needs far more $\because$ melting only loosens the lattice (molecules stay close), whereas vaporizing must overcome nearly all intermolecular attractions to separate molecules completely.

360. Ionic solid $\because$ its composition implies a very high melting point (2+/2− ions), brittleness, and conduction only when molten.

361. Coordination $= 4:4$ $\because$ each cation is surrounded by 4 anions and each anion by 4 cations in the zinc blende (ZnS) structure.

362. Yes, Tyndall effect is observed. Colloid (emulsion) $\because$ fat globules 1–1000 nm scatter light; Tyndall effect observed.

363. $i = \dfrac{\Delta T}{K_f m} = \dfrac{0.465}{1.86 \times 0.250} = 1.00$; the solute is a nonelectrolyte that stays as whole molecules, so $i = 1$ as expected.

364. $K_{\text{H}} = C/P = 1.3 \times 10^{-3}/1.0\,\text{atm} = 1.3 \times 10^{-3}\,\text{mol/(L·atm)}$; $P = \dfrac{C}{K_{\text{H}}} = \dfrac{6.5 \times 10^{-3}\,\text{mol/L}}{1.3 \times 10^{-3}\,\text{mol/(L·atm)}} = 5.0\,\text{atm}$

365. Hydrogen bonding $\because$ N–H and O–H groups let the two molecules hydrogen-bond to each other.

366. $m = \dfrac{\Delta T_b}{iK_b} = \dfrac{0.256}{1 \times 0.512} = 0.500\,\text{mol/kg}$; $n = m \times m_{\text{solvent}} = 0.500 \times 0.100\,\text{kg} = 0.0500\,\text{mol}$; $m_{\text{solute}} = 0.0500\,\text{mol} \times 342.3\,\text{g/mol} = 17.1\,\text{g}$

367. 1 $\because$ 3 N–H bonds (donor) + 1 lone pair on N (acceptor).

368. Yes $\because$ ammonia forms hydrogen bonds with water and partially ionizes to NH_4^+ and OH^-.

369. $T_2 = \left(\dfrac{1}{T_1} - \dfrac{R\ln(P_2/P_1)}{\Delta H_{\text{vap}}}\right)^{-1} = 354.4\,\text{K} = 81.3°\text{C}$ $\because$ boiling occurs when the vapor pressure equals the external pressure, and a lower external pressure lowers the boiling point.

370. Water flows from the 0.10 M solution into the 0.25 M solution. $\because$ water moves from lower solute concentration to higher solute concentration.

371. Endothermic ($\Delta H_{\text{sol}}+ = 10.9\,\text{kJ/mol}$) $\therefore$ solution temperature will decrease. $\because$ hydrogen bonding with water is slightly less favorable than solute-solute H-bonds.

372. 52.4% $\because$ 1 sphere of radius r = a/2; sphere vol = $(4/3)\pi r^3$; cell vol = a^3; efficiency = $(4/3)\pi(a/2)^3/a^3 = \pi/6 \approx 52.4\%$.

373. $i = \dfrac{\Delta T}{K_b m} = \dfrac{0.0691}{0.512 \times 0.0500} = 2.70$; the solute is dissociating incompletely: the measured $i = 2.70$ falls short of the ideal 3 because ion pairing keeps some ions associated in solution.

374. $\chi_{H_2O} = \dfrac{9.99}{9.99 + 0.899} = 0.917$; $P = 0.917 \times 23.8\,\text{mmHg} = 21.8\,\text{mmHg}$; $\Delta P = 1.96\,\text{mmHg}$ lowering.

375. Molecular solid (CO_2) $\because$ molecular crystal: neutral CO_2 molecules; very low melting point due to weak LDF.

376. $\Delta H_{\text{sol}} = \Delta H_{\text{lattice}} + \Delta H_{\text{hyd}} = +604 + (-567) = +33.3\,\text{kJ/mol}$; endothermic $\because$ $\Delta H_{\text{sol}} > 0$.

377. $i = \dfrac{\Delta T}{K_b m} = \dfrac{0.0870}{0.512 \times 0.100} = 1.70$; the solute is dissociating incompletely: the measured $i = 1.70$ falls short of the ideal 2 because ion pairing keeps some ions associated in solution.

378. Cubic holes, coordination number 8 $\because$ $r_+/r_- = 167/181 = 0.92$ lies in the 0.732–1.0 range, which geometrically favors cubic coordination.

379. $\Delta H_{\text{fus}} = \Delta H_{\text{sub}} - \Delta H_{\text{vap}} = 62.36 - 43.3 = 19.06\,\text{kJ/mol}$

380. $m = \dfrac{0.171\,\text{mol}}{0.500\,\text{kg}} = 0.342\,\text{mol/kg}$; $\Delta T_f = iK_f m = 2 \times 1.86 \times 0.342 = 1.27\,°\text{C}$; $T_f = 0.00 - 1.27 = -1.27\,°\text{C}$

381. London dispersion $\because$ nonpolar symmetric molecule.

382. $T_2 = \left(\frac{1}{T_1} - \frac{R\ln(P_2/P_1)}{\Delta H_{vap}}\right)^{-1} = 366.9\,\text{K} = 93.8°\text{C}$ $\because$ boiling occurs when the vapor pressure equals the external pressure, and a lower external pressure lowers the boiling point.

383. The liquid is vaporizing; use $q = n\Delta H_{vap}$ $\because$ the added heat converts liquid to gas at constant temperature, raising potential energy by breaking intermolecular forces rather than raising kinetic energy.

384. Saturated $\because$ rate of dissolution equals rate of crystallization.

385. 1 $\because$ 1 O–H donor + 2 lone pairs on O acceptors.

386. Liquid is being warmed; use $q = mc\Delta T$ $\because$ within a single phase the added heat increases molecular kinetic energy, so the temperature rises steadily.

387. Saturated $\because$ rate of dissolution equals rate of crystallization.

388. $T_2 = \left(\frac{1}{T_1} - \frac{R\ln(P_2/P_1)}{\Delta H_{vap}}\right)^{-1} = 345.3\,\text{K} = 72.2°\text{C}$ $\because$ boiling occurs when the vapor pressure equals the external pressure, and a lower external pressure lowers the boiling point.

389. True solution $\because$ particle size is less than 1 nm; true solutions are optically clear.

390. Hard-water Ca^{2+} and Mg^{2+} ions precipitate soap's carboxylate heads as insoluble scum, but a detergent's sulfonate/sulfate heads stay soluble with those ions, so it keeps forming micelles.

391. No layers form $\because$ acetic acid is polar (hydrogen-bonding) like water and hydrogen-bonds with it, so the two are miscible; density is irrelevant once the liquids mix.

392. Warming the ice to 0°C uses 4.95 kJ; the remaining heat only partially melts it $\therefore$ an ice–water mixture at 0°C, with 50.2 g melted.

393. Ethanol $\because$ ethanol hydrogen-bonds whereas hexane has only London forces; stronger intermolecular forces raise surface tension.

394. $C_3H_8O_3 > C_2H_5OH > H_2O > C_6H_{14}$ $\because$ viscosity $\propto$ IMF strength and molecular complexity; $C_3H_8O_3$: extensive H-bonding network across 3 –OH groups; C_2H_5OH: H-bonding; slightly less than water due to hydrophobic tail; H_2O: strong H-bonding but small molecule; C_6H_{14}: LDF only; low viscosity.

395. Endothermic ($\Delta H_{sol}+ = 35.4\,\text{kJ/mol}$) $\therefore$ solution temperature will decrease. $\because$ large positive ΔHsol; entropy increase drives dissolution despite endothermicity.

396. $C_5H_{11}COOH > C_3H_7COOH > CH_3COOH > HCOOH$ $\because$ a longer hydrocarbon tail matches hexane's dispersion forces better; the polar $-$COOH head is what resists dissolving, so longer chains dissolve more readily.

397. Solubility increases by a factor of 3 $\because$ $C = K_HP$; $C_1 = 3.0 \times 10^{-4}$ mol/L, $C_2 = 9.2 \times 10^{-4}$ mol/L; $C_2/C_1 = 1.5/0.5 = 3$.

398. $m = \frac{\Delta T_b}{iK_b} = \frac{0.256}{1 \times 0.512} = 0.500\,\text{mol/kg}$; $n = m \times m_{solvent} = 0.500 \times 1.00\,\text{kg} = 0.500\,\text{mol}$; $m_{solute} = 0.500\,\text{mol} \times 342.3\,\text{g/mol} = 171\,\text{g}$

399. Sol $\because$ it is a solid dispersed in a liquid.

400. Concave (curves up at the walls) $\because$ adhesion of water to glass exceeds cohesion within the water, so the liquid climbs the walls.

401. $C_6H_{12}O_6$ $(i = 1) <$ NaCl $(i = 2) < CaCl_2$ $(i = 3)$ $\because$ all colligative properties $\propto im$; larger $i \rightarrow$ greater effect. $C_6H_{12}O_6$: nonelectrolyte; 1 particle per formula unit; NaCl: strong electrolyte; $Na^+ + Cl^- \rightarrow$ 2 particles; $CaCl_2$: strong electrolyte; $Ca^{2+} + 2Cl^-$ $\rightarrow$ 3 particles.

402. 1 $\because$ 8 corner atoms $\times$ 1/8 each.

403. Yes, Tyndall effect is observed. Colloid (aerosol) $\because$ solid carbon particles 1–1000 nm scatter light; Tyndall effect visible.

404. Molecular solid $\because$ its composition implies a very low melting point, softness, and no electrical conduction.

405. $M = \frac{\rho \cdot N_A \cdot a^3}{Z} = \frac{8.94 \times 6.022 \times 10^{23} \times (361.49 \times 10^{-10})^3}{4} = 63.5\,\text{g/mol}$ $\therefore$ the metal is copper (Cu).

406. $\Pi = iMRT = 2 \times 0.154 \times 0.08206 \times 298\,\text{K} = 7.54\,\text{atm}$; minimum applied pressure must exceed 7.54 atm $\because$ reverse osmosis requires $P_{applied} > \Pi$.

407. Solid foam $\because$ it is a gas dispersed in a solid.

408. syrup $> C_3H_8O_3 > C_2H_5OH >$ gasoline $\because$ viscosity $\propto$ IMF strength and molecular complexity; syrup: very strong intermolecular attractions between sugar molecules; $C_3H_8O_3$: three hydroxyl groups create extensive H-bonding; C_2H_5OH: moderate H-bonding; gasoline: mostly nonpolar hydrocarbons with weak LDF.

409. $C_{12}H_{22}O_{11}$ $(i = 1) <$ KCl $(i = 2) < AlCl_3$ $(i = 4)$ $\because$ all colligative properties $\propto im$; larger $i \rightarrow$ greater effect. $C_{12}H_{22}O_{11}$: nonelectrolyte; 1 particle; KCl: strong electrolyte; $K^+ + Cl^- \rightarrow$ 2 particles; $AlCl_3$: strong electrolyte; $Al^{3+} + 3Cl^- \rightarrow$ 4 particles.

410. $\Pi = iMRT = 1 \times 0.500 \times 0.08206 \times 298\,\text{K} = 12.2\,\text{atm}$

411. $K_H = C/P = 6.1 \times 10^{-4}/1.0\,\text{atm} = 6.1 \times 10^{-4}\,\text{mol/(L·atm)}$; $P = \dfrac{C}{K_H} = \dfrac{2.4 \times 10^{-3}\,\text{mol/L}}{6.1 \times 10^{-4}\,\text{mol/(L·atm)}} = 4.0\,\text{atm}$

412. $n_{\text{solute}} = \dfrac{50.0}{62.07} = 0.806\,\text{mol}$. $m = \dfrac{0.806\,\text{mol}}{0.200\,\text{kg}} = 4.03\,\text{mol/kg}$

413. $\Delta H_{\text{vap}} = \dfrac{-R\ln(P_2/P_1)}{1/T_2 - 1/T_1} = 31.3\,\text{kJ/mol}$

414. Ionic solid (NaCl) ∵ ions held by electrostatic forces; ions mobile only when molten.

415. $M = \dfrac{\rho \cdot N_A \cdot a^3}{Z} = \dfrac{2.7 \times 6.022 \times 10^{23} \times (404.95 \times 10^{-10})^3}{4} = 27.0\,\text{g/mol}$ ∴ the metal is aluminum (Al).

416. Ideal behavior ∵ both are similar nonpolar hydrocarbons with comparable London dispersion forces → ideal behavior.

417. Hydrogen bonding ∵ N–H and O–H groups let the two molecules hydrogen-bond to each other.

418. Boiling needs far more ∵ melting only loosens the lattice (molecules stay close), whereas vaporizing must overcome nearly all intermolecular attractions to separate molecules completely.

419. 1 ∵ 8 corner atoms × 1/8 each.

420. $m = \dfrac{\Delta T_b}{iK_b} = \dfrac{0.512}{1 \times 0.512} = 1.00\,\text{mol/kg}$; $n = m \times m_{\text{solvent}} = 1.00 \times 1.00\,\text{kg} = 1.00\,\text{mol}$; $m_{\text{solute}} = 1.00\,\text{mol} \times 180.16\,\text{g/mol} = 180.\,\text{g}$

421. CH_4 (-164°C) < C_2H_6 (-89°C) < C_3H_8 (-42°C) < C_4H_{10} (-1°C). ∵ These molecules are nonpolar, so only London dispersion forces are significant. Increasing molecular size increases polarizability, strengthening intermolecular attractions and increasing the boiling point.

422. $n_{\text{solute}} = \dfrac{25.0}{180.16} = 0.139\,\text{mol}$; $n_{\text{water}} = \dfrac{500.0}{18.015} = 27.8\,\text{mol}$. $\chi_{\text{solute}} = \dfrac{0.139}{0.139 + 27.8} = 4.97 \times 10^{-3}$

423. $F = 1 - 1 + 2 = 2$. ∵ Both temperature and pressure may be varied independently because a single phase imposes no phase-equilibrium constraint between them.

424. London dispersion ∵ nonpolar homonuclear diatomic.

425. $\chi_{H_2O} = \dfrac{11.1}{11.1 + 0.0526} = 0.995$; $P = 0.995 \times 355.1\,\text{mmHg} = 353\,\text{mmHg}$; $\Delta P = 1.67\,\text{mmHg}$ lowering.

426. $\rho = \dfrac{ZM}{N_A a^3} = \dfrac{4 \times 107.868}{6.022 \times 10^{23} \times (408.53 \times 10^{-10})^3} = 10.5\,\text{g/cm}^3$. ∵ density is determined by the mass contained in one unit cell divided by the unit-cell volume. The required information is the molar mass, the number of atoms per unit cell (obtained directly or from the crystal structure), and the unit-cell edge length; the remaining data do not affect the calculation.

427. There is no net flow of water. ∵ both solutions have the same solute concentration and osmotic pressure.

428. A face-centered cubic (FCC) unit cell, filling 74% of space ∵ ABCABC stacking is geometrically identical to the FCC lattice.

429. $n_{eth} = \dfrac{18.4\,\text{g}}{46.07} = 0.400\,\text{mol}$; $n_{H_2O} = \dfrac{100.\,\text{g}}{18.015} = 5.55\,\text{mol}$; $\chi_{eth} = 0.0672$; $\chi_{H_2O} = 0.933$; $\chi_{eth} + \chi_{H_2O} = 1.00$ ✓

430. $m = \dfrac{q}{\Delta H} = \dfrac{35.0 \times 10^3\,\text{J}}{571\,\text{J/g}} = 61.3\,\text{g}$

431. Two layers form and water is on top ∵ carbon tetrachloride is nonpolar and cannot hydrogen-bond with water, so they are immiscible; the less dense liquid floats, and density fixes only the stacking order, not whether they mix.

432. $AlCl_3$ ∴ greatest freezing point depression ∵ largest effective molality. CH_4N_2O: $i \times m = 1 \times \dfrac{1.00}{60.06} = 0.0167\,\text{mol/kg}$; KCl: $i \times m = 2 \times \dfrac{1.00}{74.55} = 0.0268\,\text{mol/kg}$; $AlCl_3$: $i \times m = 4 \times \dfrac{1.00}{133.34} = 0.0300\,\text{mol/kg}$.

433. $m = \dfrac{0.125\,\text{mol}}{0.100\,\text{kg}} = 1.25\,\text{mol/kg}$; $\Delta T_f = iK_f m = 2 \times 1.86 \times 1.25 = 4.63\,°\text{C}$; $T_f = 0.00 - 4.63 = -4.63\,°\text{C}$

434. London dispersion ∵ nonpolar homonuclear diatomic; no permanent dipole.

435. $\Delta H_{\text{vap}} = \dfrac{-R\ln(P_2/P_1)}{1/T_2 - 1/T_1} = 26.5\,\text{kJ/mol}$

436. $n_{ace} = \dfrac{12.8\,\text{g}}{58.08} = 0.221\,\text{mol}$; $n_{H_2O} = \dfrac{50.0\,\text{g}}{18.015} = 2.78\,\text{mol}$; $\chi_{ace} = 0.0738$; $\chi_{H_2O} = 0.926$; $\chi_{ace} + \chi_{H_2O} = 1.00$ ✓

437. $U = \Delta H_f - \Delta H_{\text{sub}} - IE_1 - \frac{1}{2}D - EA = -617 - 159 - 520 - 79 - (-328) = -1047\,\text{kJ/mol}$

438. Metallic solid (Au) ∵ delocalized electrons (sea of electrons) allow electrical conductivity and confer malleability.

439. $U = \Delta H_{\mathrm{f}} - \Delta H_{\mathrm{sub}} - IE_1 - \frac{1}{2}D - EA = -617 - 159 - 520 - 79 - (-328) = -1047\,\mathrm{kJ/mol}$

440. It will sublime ∵ the external pressure (2.527 atm) is below the triple-point pressure (5.11 atm), so no liquid is stable and the solid converts directly to gas.

441. $m = \dfrac{0.333\,\mathrm{mol}}{0.250\,\mathrm{kg}} = 1.33\,\mathrm{mol/kg}$; $\Delta T_f = iK_f m = 1 \times 1.86 \times 1.33 = 2.48\,°\mathrm{C}$; $T_f = 0.00 - 2.48 = -2.48\,°\mathrm{C}$

442. LiI (-757 kJ/mol) < LiBr (-807 kJ/mol) < LiCl (-853 kJ/mol) < LiF (-1037 kJ/mol) ∵ lattice energy becomes more negative with higher ion charges and smaller ionic radii (Coulomb's law: $U \propto \dfrac{q_+q_-}{r}$).

443. 52.4% ∵ 1 sphere of radius r = a/2; sphere vol = (4/3)πr^3; cell vol = a^3; efficiency = (4/3)π(a/2)^3/a^3 = π/6 ≈ 52.4%.

444. $\Delta H_{\mathrm{vap}} = 394\,\mathrm{J/g} \times 78.114\,\mathrm{g/mol} \times \dfrac{1\,\mathrm{kJ}}{1000\,\mathrm{J}} = 30.8\,\mathrm{kJ/mol}$

445. Negative deviation from Raoult's law ∵ extensive hydrogen bonding between methanol and water strengthens intermolecular attractions → lower vapor pressure than ideal.

446. Coordination = 8 : 4 ∵ each cation is surrounded by 8 anions and each anion by 4 cations in the fluorite (CaF2) structure.

447. Solubility negligible change ∵ pressure has essentially no effect on solubility of solids/liquids.

448. Water flows from pure water into the NaCl solution. ∵ water moves from lower solute concentration (pure water) to higher solute concentration (NaCl) through a semipermeable membrane.

449. A quartz crystal is crystalline (a regular, repeating lattice); rubber is amorphous (no long-range order) ∵ crystallinity is defined by long-range periodic order.

450. $n_{eth} = \dfrac{12.8\,\mathrm{g}}{46.07} = 0.279\,\mathrm{mol}$; $n_{\mathrm{H_2O}} = \dfrac{90.0\,\mathrm{g}}{18.015} = 5.00\,\mathrm{mol}$; $\chi_{eth} = 0.0528$; $\chi_{\mathrm{H_2O}} = 0.947$; $\chi_{eth} + \chi_{\mathrm{H_2O}} = 1.00$ ✓

451. It will melt ∵ the external pressure (0.025 atm) is above the triple-point pressure (0.006 atm), so a stable liquid phase exists between solid and gas.

452. Melting all 37.6 g uses 12.6 kJ; the remaining heat warms the water ∴ liquid water at $T = \dfrac{q - mH_{\mathrm{fus}}}{mc} = 32.7°\mathrm{C}$.

453. It is the equilibrium separation between molecules ∵ at the minimum the net force is zero (attraction balances repulsion), and molecules in a solid vibrate about this distance.

454. Water flows from the 0.05 M solution into the 0.20 M solution. ∵ water moves from lower solute concentration to higher solute concentration.

455. Mole fraction. ∵ it is moles of one component over total moles, it is dimensionless and temperature-independent, which makes it ideal for vapor-pressure (Raoult's law) calculations.

456. Vaporization (liquid → gas) ∵ the path crosses only the boiling point (−183 °C); it starts as liquid (mp −218 °C).

457. Per 1 L of solution: mass solution $= 1.02 \times 10^3\,\mathrm{g}$; mass solute $= 0.500\,\mathrm{mol} \times 180.16\,\mathrm{g/mol} = 90.1\,\mathrm{g}$; mass solvent $= 1.02 \times 10^3\,\mathrm{g} - 90.1\,\mathrm{g} = 928\,\mathrm{g}$; $m = \dfrac{0.500\,\mathrm{mol}}{0.928\,\mathrm{kg}} = 0.539\,\mathrm{mol/kg}$

458. Water has the higher boiling point. ∵ its hydrogen atoms are bonded directly to oxygen, neighboring molecules can form hydrogen bonds. These intermolecular attractions are significantly stronger than the intermolecular forces in the other substance, so more energy is required to separate the molecules into the gas phase, producing the higher boiling point.

459. $\rho = \dfrac{Z \cdot M}{N_A \cdot a^3} = \dfrac{4 \times 58.443}{6.022 \times 10^{23} \times (564.0 \times 10^{-10})^3} = 2.16\,\mathrm{g/mL}$

460. Kno$_3$ crystallizes from solution on cooling. ∵ KNO_3 solubility increases with T; cooling creates supersaturation → crystallization.

461. Melting destroys long-range crystal order, which cannot vanish continuously ∵ liquid and gas differ only in density and merge smoothly, but a crystal's long-range order is a symmetry that must break abruptly.

462. Yes, Tyndall effect is observed. Colloid (gel) ∵ polymer chains 1–1000 nm scatter light.

463. $M = \dfrac{\rho \cdot N_A \cdot a^3}{Z} = \dfrac{2.7 \times 6.022 \times 10^{23} \times (404.95 \times 10^{-10})^3}{4} = 27.0\,\mathrm{g/mol}$ ∴ the metal is aluminum (Al).

464. Dissolved n$_2$ comes out of solution (decompression sickness). ∵ reduced pressure at surface lowers N_2 solubility (Henry's law); dissolved N_2 forms bubbles.

465. N-butane ∵ with identical formula and mass, the more elongated shape has greater surface contact area, producing stronger London dispersion forces and a higher boiling point.

466. Strong electrolyte ∵ ionic compound; dissociates completely: $Na^+ + Cl^-$.

467. $q = mc\Delta T = 39.8 \times 2.09 \times 22 = 1.83\,\mathrm{kJ}$

468. Vaporization (liquid $\rightarrow$ gas) $\because$ the path crosses only the boiling point (-183 °C); it starts as liquid (mp -218 °C).

469. Ammonia has the higher boiling point. $\because$ its hydrogen atoms are bonded directly to nitrogen, neighboring molecules can form hydrogen bonds. These intermolecular attractions are significantly stronger than the intermolecular forces in the other substance, so more energy is required to separate the molecules into the gas phase, producing the higher boiling point.

470. $i = \dfrac{\Delta T}{K_f m} = \dfrac{0.558}{1.86 \times 0.100} = 3.00$; the solute is dissociating completely into 3 ions, matching the ideal $i = 3$.

471. Endothermic ($\Delta H_{\mathrm{sol}} + = 25.7\,\mathrm{kJ/mol}$) $\therefore$ solution temperature will decrease. $\because$ lattice energy > solvation energy; system absorbs heat; entropy increase drives dissolution.

472. Ionic solid (MgO) $\because$ ionic crystal with high melting point due to +2/-2 charges; non-conducting as solid.

473. $d = \dfrac{n\lambda}{2\sin\theta} = \dfrac{1 \times 154}{2\sin(10.3°)} = 431\,\mathrm{pm}$. $\because$ waves reflected from adjacent crystal planes travel different distances. A diffraction maximum occurs only when that path difference equals an integer multiple of the wavelength, so constructive interference requires $n\lambda = 2d\sin\theta$.

474. $C_3H_7NH_2$ > $C_4H_9NH_2$ > $C_6H_{13}NH_2$ > $C_8H_{17}NH_2$ $\because$ all share the same H-bonding $-NH_2$ head, so the nonpolar chain decides solubility: a longer hydrocarbon tail disrupts water's H-bond network more and lowers solubility.

475. It will sublime $\because$ the external pressure (0.0051 atm) is below the triple-point pressure (0.017 atm), so no liquid is stable and the solid converts directly to gas.

476. 52.4% $\because$ 1 sphere of radius r = a/2; sphere vol = (4/3)πr^3; cell vol = a^3; efficiency = (4/3)π(a/2)^3/a^3 = π/6 $\approx$ 52.4%.

477. Octahedral holes, coordination number 6 $\because$ $r_+/r_- = 102/181 = 0.56$ lies in the 0.414–0.732 range, which geometrically favors octahedral coordination.

478. $i = 3$; $m = \dfrac{0.0901\,\mathrm{mol}}{0.200\,\mathrm{kg}} = 0.450\,\mathrm{mol/kg}$; $\Delta T_b = iK_b m = 3 \times 0.512 \times 0.450 = 0.692\,°\mathrm{C}$; $T_b = 100.0 + 0.692 = 101\,°\mathrm{C}$

479. $U = \Delta H_{\mathrm{f}} - \Delta H_{\mathrm{sub}} - IE_1 - \frac{1}{2}D - EA = -437 - 89 - 419 - 121 - (-349) = -717\,\mathrm{kJ/mol}$

480. Yes $\because$ ion-dipole forces between Na^+/Cl^- and H_2O overcome the lattice energy.

481. $q = m\Delta H_{\mathrm{vap}} = 21.1 \times 2260 = 47.7\,\mathrm{kJ}$

482. $CaCl_2$: $i \times M = 3 \times 0.100\,\mathrm{M} = 0.300\,\mathrm{osmol/L}$ (isotonic); CH_4N_2O: $i \times M = 1 \times 0.250\,\mathrm{M} = 0.250\,\mathrm{osmol/L}$ (hypotonic); NaCl: $i \times M = 2 \times 0.150\,\mathrm{M} = 0.300\,\mathrm{osmol/L}$ (isotonic). $\because$ tonicity depends on the total particle concentration $i \times M$, not on molarity alone.

483. Cubic holes, coordination number 8 $\because$ $r_+/r_- = 167/181 = 0.92$ lies in the 0.732–1.0 range, which geometrically favors cubic coordination.

484. Molecular solid (I_2) $\because$ molecular crystal: neutral I_2 molecules held by LDF; no free charges.

485. $\Delta H_{\mathrm{sub}} = \Delta H_{\mathrm{fus}} + \Delta H_{\mathrm{vap}} = 6.01 + 40.7 = 46.71\,\mathrm{kJ/mol}$

486. Endothermic ($\Delta H_{\mathrm{sol}} + = 25.7\,\mathrm{kJ/mol}$) $\therefore$ solution temperature will decrease. $\because$ lattice energy > solvation energy; system absorbs heat; entropy increase drives dissolution.

487. $\Delta H_{\mathrm{sol}} = \Delta H_{\mathrm{lattice}} + \Delta H_{\mathrm{hyd}} = +853 + (-890) = -37.0\,\mathrm{kJ/mol}$; exothermic $\because$ $\Delta H_{\mathrm{sol}} < 0$.

488. $C = K_{\mathrm{H}} P = 1.3 \times 10^{-3} \times 4.0\,\mathrm{atm} = 5.2 \times 10^{-3}\,\mathrm{mol/L}$

489. It has liquid-like density but gas-like viscosity $\because$ high density supplies frequent solute contacts (dissolving power) while weak intermolecular resistance permits rapid, gas-like diffusion.

490. $\Pi = iMRT = 1 \times 0.0500 \times 0.08206 \times 298\,\mathrm{K} = 1.22\,\mathrm{atm}$; minimum applied pressure must exceed 1.22 atm $\because$ reverse osmosis requires $P_{\mathrm{applied}} > \Pi$.

491. Emulsion; e.g., mayonnaise. $\because$ a liquid-in-liquid dispersion defines this colloid type.

492. No — spontaneity is not the same as exothermicity. $\because$ dissolving scatters the ordered crystal lattice into freely moving hydrated ions, the large entropy increase outweighs the endothermic ΔH, so the process proceeds even while it cools the surroundings.

493. $HOCH_2CH_2OH$ > C_3H_7OH > CH_3OH > C_5H_{12} $\because$ viscosity $\propto$ IMF strength and molecular complexity; $HOCH_2CH_2OH$: 2 –OH groups allow strong H-bonding; C_3H_7OH: H-bonding and larger electron cloud; CH_3OH: H-bonding but very small molecule; C_5H_{12}: weak LDF only.

494. C_4H_9OH > $C_5H_{11}OH$ > $C_6H_{13}OH$ > $C_8H_{17}OH$ $\because$ all share the same H-bonding $-OH$ head, so the nonpolar chain decides solubility: a longer hydrocarbon tail disrupts water's H-bond network more and lowers solubility.

495. 2 $\because$ 2 O–H bonds (donor) + 2 lone pairs on O (acceptor).

496. KF (-821 kJ/mol) < NaF (-923 kJ/mol) < LiF (-1037 kJ/mol) $\because$ lattice energy becomes more negative with higher ion charges and smaller ionic radii (Coulomb's law: $U \propto \frac{q_+q_-}{r}$).

497. Na_2CO_3 $\therefore$ greatest freezing point depression $\because$ largest effective molality. $C_6H_{12}O_6$: $i \times m = 1 \times \frac{1.00}{180.16} = 5.55 \times 10^{-3}\,\text{mol/kg}$; $Ca(NO_3)_2$: $i \times m = 3 \times \frac{1.00}{164.1} = 0.0183\,\text{mol/kg}$; Na_2CO_3: $i \times m = 3 \times \frac{1.00}{105.99} = 0.0283\,\text{mol/kg}$.

498. $m = \frac{0.0584\,\text{mol}}{0.200\,\text{kg}} = 0.292\,\text{mol/kg}$; $\Delta T_b = iK_b m = 1 \times 0.512 \times 0.292 = 0.150\,^\circ\text{C}$; $T_b = 100.0 + 0.150 = 100.\,^\circ\text{C}$

499. Glycerol $\because$ glycerol's three O–H groups give far more hydrogen bonding than ethanol's one; stronger intermolecular forces increase viscosity.

500. Water flows from pure water into the glucose solution. $\because$ water moves toward the solution with higher solute concentration.